

The Impact of Generative AI on Information Processing: Evidence from the Ban of ChatGPT in Italy

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Abstract

This paper explores how the emergence of generative artificial intelligence is reshaping the information environment in capital markets. Leveraging an unexpected ban on ChatGPT in Italy, we examine its impact on the information processing capabilities of market participants. We employ metrics for AI-generated text detection to show that the ban coincides with decreased AI usage by domestic financial analysts and fewer earnings forecasts issued relative to foreign analysts covering the same firm. The negative effects are more pronounced among analysts whose pre-ban reports are more consistent with AI use or analysts with a technical background. The ban also diminishes forecast accuracy, increases reliance on industry-specific information, and reduces information efficiency. Furthermore, investor reactions to earnings announcements become more pronounced, and bid–ask spreads widen, reflecting lower market efficiency.

Keywords: *AI, generative, ChatGPT, analyst, information processing, market efficiency.*

JEL Codes: O00, O33, O38, E44, E65, G10, G14.

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I. Introduction

The advent of generative AI has transformed access to computer-assisted productivity tools. Leading technologies, such as Google’s Gemini and OpenAI’s ChatGPT, have equipped individuals worldwide with unprecedented information processing capabilities. As Lawrence H. Summers, former US Treasury Secretary, stated on Bloomberg TV on December 9, 2022, “ChatGPT is a development on par with the printing press, electricity, and even the wheel and fire.” Providing capital market participants with advanced information processing tools can have profound implications for the informational efficiency of the capital market (Sims, 2003; Li, 2008; Dong et al., 2016; Blankespoor et al., 2019; Blankespoor et al., 2020). Within the rational expectations framework, reducing the cost of accessing and processing information should enhance market efficiency (see Verrecchia, 1982, p. 1426).

This paper examines whether adopting new tools that reduce informational friction actually improves information flows and the capital market’s informational efficiency. However, the empirical challenge is that the adoption of generative AI varies in intensity and timing across firms and individuals, and utilization correlates with many other economic characteristics. To overcome this challenge, we leverage a setting in which, on March 31, 2023, Italian regulators banned ChatGPT within its geographic borders due to concerns about consent and personal data protection, and access to servers from IP addresses from Italy was blocked. After nearly a month of discussions and improvements, Italian authorities re-authorized the service on April 28, 2023.¹

¹ A regulatory agency enforced the ban to comply with privacy laws, which were plausibly exogenous to factors like perceived productivity gains or lobbying by affected groups. The ban came without advance warning from Italian regulators or ongoing discussions with OpenAI, making it a surprise to most observers. At the time, there were no complete alternatives to ChatGPT. Google’s Bard (now Gemini) launched in Europe in late June, and Meta’s AI tools were less versatile, while Bing AI, launched on February 7, 2023, focused on search capabilities and had limited features. Microsoft 365 Copilot, announced on March 16, 2023, began rolling out in November 2023.

The ban's abrupt timing and geographic scope have allowed us to better control for trends in adopting related technologies correlated with generative AI use. More importantly, the ban took effect after ChatGPT was already widely used in Italy, but before other AI options were available.

We focus on the effect of the ban on the work of financial analysts for several reasons. First, analysts play a well-documented role in information intermediation and in shaping the capital market's informational efficiency (Womack, 1996; Healy and Palepu, 2001; Bradshaw, 2011; Bradshaw et al., 2016; Kothari et al., 2016).² Analysts can use prompts to guide ChatGPT in efficiently preparing and analyzing information. Second, analysts' job performance and outputs are empirically measurable, as they provide metrics that capture technological adoption and information processing activities (Li et al., 2024). Third, the adoption of ChatGPT is influenced by technological savviness, with financial analysts with technical backgrounds being more likely to adopt it (e.g., Bick et al., 2024). By collecting demographic information on individual analysts, such as education and location, we are able to assess the differential impacts of ChatGPT adoption across various groups.

We investigate analysts' company reports to explore whether the ban might have reduced their use of the tool. Since not all analysts use AI technologies, and some may have found workarounds, such as personal virtual private networks (VPNs), the ban was unlikely to impact everyone equally. Nevertheless, the linguistic cues within the reports may indicate the likelihood of ChatGPT usage. This is particularly relevant in our setting, where most domestic analysts are non-native English speakers writing in English. Thus, reports edited with ChatGPT (pre-ban) could differ from those

² We consulted several sell-side financial analysts, including a local Italian analyst, about their use of ChatGPT. Some began using it in early 2023, shortly after the launch of ChatGPT 3.5. They utilized its search and summarization capabilities to gather preliminary information on unfamiliar sectors or companies, focusing on growth drivers and major players. In addition, they used ChatGPT to process information from public repositories. While they recognized ChatGPT as a valuable tool for daily tasks, they also expressed concerns about its potential impact on future careers.

composed without ChatGPT (during the ban). In light of its focus on reports, this analysis represents an initial step toward measuring the adoption of new technologies among financial analysts, which is typically unobservable and rarely examined in prior studies.

To conduct our analysis, we rely on widely used text-based metrics to aid in AI-generated text detection (see Appendix B). To assess differences in ChatGPT usage between domestic and foreign analysts, we use analysts' office addresses and email domains to determine their locations. Reports authored by domestic analysts exhibit a lower likelihood of AI-generated content during the ban. However, after the restoration of ChatGPT, we do not observe significant differences in AI-generated content between domestic and foreign analysts. To refine our analysis, we collect data from resumes to identify analysts majoring in technical subjects (e.g., STEM majors) during undergraduate or postgraduate studies (Frank et al., 2023). These "tech analysts" are presumably more inclined to use new technology and are therefore more likely to be impacted by the ban. Indeed, the change in text-based metrics indicating AI use in the reports of domestic analysts was more pronounced among "tech analysts." Overall, these findings collectively indicate that the ban diminished financial analysts' use of AI.

Next, we examine how reduced ChatGPT usage affects the production of information by financial analysts. To do this, we compare domestic analysts' intermediation activities with those of foreign analysts covering the same firms. Our results show that during the ban, the number of forecasts by domestic analysts decreases by 21% of the sample standard deviation compared to those produced by foreign analysts, suggesting that the ban impairs the analysts' information intermediation. We also find that the reduction in information production was reversed in the months after ChatGPT was reactivated. We perform two cross-sectional analyses examining the ban's impact on different groups of analysts, differentiated by their educational background and

likely prior usage of ChatGPT. The observed decrease in the number of forecasts during the ban is predominantly driven by domestic analysts with a technical major or whose reports are more consistent with AI-generated content.

Enhancing analysts' processing capacity can increase both the quantity and quality of their forecasts. Documenting the significant impact of the ChatGPT ban on forecast accuracy suggests that analysts use ChatGPT for complex tasks, such as data input and prompt-based information processing, rather than merely for writing (Kim et al., 2024). We find consistent evidence that the ban reduced forecast accuracy, especially among analysts with high technological expertise or those for whom prior AI usage was detected.

We additionally analyze industry-level information in analyst reports, as this type of information typically has lower processing costs and is prioritized when information processing capacity is limited (Peng, 2005; Peng and Xiong, 2006). Using machine learning to identify industry-related keywords (Li et al., 2021), we measure the proportion of such information in the analyst reports. Our analysis shows a significant increase in reliance on industry-level information among domestic analysts during the ban, indicating that reduced processing capacity may lead to a greater focus on easily processed information.

We also examine how reduced activity by financial analysts affects the informational efficiency of capital markets (Grossman and Stiglitz, 1980; Verrecchia, 1982; Blankespoor et al., 2020). In this case, we anticipate that the ban would be found to have limited the analysts' information production, leading investors to depend more on public information from earnings announcements and thereby increasing their reaction to these announcements (Kyle, 1985; Back, 1992; Ball and Shivakumar, 2008). We find consistent evidence for this hypothesis.

Furthermore, our results show increased bid–ask spreads for Italian stocks during the ban,

which suggests that a widely used new technology, such as ChatGPT, may “level the playing field” by facilitating dissemination to investors with an informational disadvantage (Verrecchia, 1982; Kyle, 1985). Further analyses reveal that firms with lower analyst coverage exhibit a more pronounced increase in bid–ask spreads, which highlights the crucial role of analysts in information intermediation. Our results suggest that the ban led to higher information processing costs and, consequently, a less informationally efficient market.

Our research contributes to multiple strands of the literature. First, our work is tied to the recent studies in finance and accounting demonstrating the capacity of AI in extracting information from unstructured sources, such as determining the sentiment of news headlines (Lopez-Lira and Tang, 2023), predicting future earnings per share (Li et al., 2023), forming economic expectations based on news articles (Bybee, 2023), improving the readability of management disclosures and conference calls (Frankel et al., 2022; Kim et al., 2023), quantifying new information conveyed by managers in conference calls (Bai et al., 2023), measuring managers’ anticipated capital expenditure (Jha et al., 2023), and determining business complexity (Bernard et al., 2023).³ Our study contributes to this stream of research by presenting timely evidence of the impact of generative AI technologies on the information processing capabilities of capital market participants.⁴

³ In addition to textual analysis, ChatGPT can potentially complete a broader range of tasks. For example, an emerging cluster of literature shows that ChatGPT can assist in auditing (Emett et al., 2023; Eulerich and Wood, 2023; Gu et al., 2023), test taking (OpenAI, 2023; Eulerich et al., 2023; Katz et al., 2023), answering common and accounting-specific questions (Wood et al., 2023; O’Leary, 2023a, 2023b), and conducting research (de Kok, 2023; Korinek, 2023).

⁴ To our knowledge, Kreitmer and Raschky’s (2023) work is the only other study that examines the impact of an exogenous shock on generative AI using the Italian ban. Specifically, they found a short-term effect of the ban on the output of coding engineers. They also document that this sophisticated user base quickly found remedial strategies, as evidenced by a recovery of coding activity after a few days and an increase in search for proxy servers (VPN). Italian businesses could, in principle, use VPN to bypass IP restrictions. This may lead to an underestimation of the treatment effect.

Second, our study adds to the large body of work examining individual analyst characteristics (Womack, 1996; Healy and Palepu, 2001; Jegadeesh et al., 2004; Bradshaw, 2011; Bradshaw et al., 2016; Li et al., 2024). In this regard, we investigate the information processing capabilities of financial analysts, providing causal evidence of how modern information technologies, particularly generative AI, affect intermediation activities. By analyzing the content of analyst reports, we identify the use of generative AI technology and evaluate its impact on report characteristics, thereby shedding light on analysts' information production.

Third, our study contributes to the information processing literature reviewed by Blankespoor et al. (2020). Specifically, this body of theoretical work suggests that limited information processing capacity hampers market participants' ability to process a comprehensive set of information, leading to informationally inefficient asset prices (Sims, 2003; Peng, 2005). We add to this literature by introducing a novel setting, the ChatGPT ban in Italy, and testing the effect of processing costs on informational efficiency.

II. Institutional Background and Conceptual Framework

2.1. Institutional Background of the ChatGPT Ban and Restoration in Italy

OpenAI is a non-profit laboratory founded in 2015 to promote AI development. The organization first released GPT-1 as open source on June 11, 2018, and GPT-2 followed on February 14, 2019 (Radford et al., 2018, 2019), to general acclaim. GPT-1 and GPT-2 were primarily designed as chatbots with capabilities for text generation rather than reliable productivity tools. On November 30, 2022, OpenAI released ChatGPT, built on the GPT-3 architecture, first to select beta testers and then to the general public on January 30, 2023. The increased size of GPT-3, approximately 100 times the size of GPT-2, made GPT-3 versatile for performing a wide range

of complex natural language tasks. GPT-4 was released on March 14, 2023, with about 1,000 times the number of parameters of GPT-3 and vastly improved capabilities to perform complex tasks (OpenAI, 2023).

By December 4, 2022, ChatGPT had already attracted over one million users. By January 2023, it had reached over 100 million users, becoming the fastest-growing consumer application ever.⁵ A March 2023 survey by Pew Research found that 14% of American adults had used ChatGPT, which rose to 18% by July. In addition, the propensity to use ChatGPT increased with the user's level of education.⁶ A global survey published by US trading platform eToro on July 11, 2023, revealed that 41% of retail investors from developed countries—including the UK, US, Germany, France, Australia, Italy, and Spain—are open to or already using AI to aid in their investment decisions.⁷

On March 20, 2023, OpenAI paused the service for four days after discovering a security vulnerability, which allowed access to other users' chat histories and personal information over a nine-hour window. Although OpenAI patched the vulnerability shortly after the breach, the Italian data protection authority (Garante per la protezione dei dati personali) determined that OpenAI had violated EU privacy laws. On March 30, Italian regulators issued an initial decision (Registro dei provvedimenti n. 112) requiring a “temporary limitation of the processing of personal data of

⁵ <https://www.theguardian.com/technology/2023/feb/02/chatgpt-100-million-users-open-ai-fastest-growing-app>

⁶ <https://www.pewresearch.org/short-reads/2023/08/28/most-americans-havent-used-chatgpt-few-think-it-will-have-a-major-impact-on-their-job/>

⁷ eToro, a brokerage platform with over 30 million users worldwide, commissioned a survey of 10,000 retail investors across 13 countries from June 2 to June 14, 2023. Respondents included self-directed or advised investors holding at least one product, such as shares or bonds, with 1,000 participants each from the UK, US, Germany, France, Australia, Italy, and Spain. The survey, published on July 11, 2023, shows that 41% of retail investors were open to or already using AI for investment decisions, with 46% saying ChatGPT saved them time on research. Sources: <https://www.etoro.com/news-and-analysis/press-releases/etoro-reports-continued-growth-with-fy22-results>; <https://www.etoro.com/en-us/news-and-analysis/latest-news/press-release/four-in-ten-u-s-retail-investors-use-or-are-open-to-using-chat-gpt-style-ai-to-help-pick-and-manage-investments>.

data subjects established in the Italian territory” (translated from Italian). They clarified this initial decision in a statement dated March 31, noting that “it [OpenAI] will have to notify the Italian SA within 20 days of the measures implemented to comply with the order, [or] a fine of up to EUR 20 million EUR or 4% of worldwide sales may be imposed.”⁸

The decision was largely unexpected. There were no enforcement actions because, under this threat, OpenAI quickly discontinued its service for IP addresses originating from Italy on March 31, 2023. The news received mixed reactions, with Deputy Prime Minister Matteo Salvini stating, “This is a work tool that many young people, many companies, many start-ups were using, and I hope that they can use it again as soon as possible, because otherwise Italy will have a gap compared to all other European countries” (Reuters, April 4, 2023). As OpenAI met with Italian regulators, it was widely expected that a reactivation would occur, but no timetable was set (AP News, “OpenAI to offer remedies to resolve Italy’s ChatGPT ban,” April 6, 2023).

On April 28, the service was reactivated after OpenAI introduced age verification to exclude users below 13 years of age and allow users to exercise their right to object to the training of models with their personal data (“ChatGPT: OpenAI reopens the platform in Italy, guaranteeing more transparency and more rights for European users and non-users,” press release, April 28, 2023). However, while the service has been uninterrupted since its reactivation, there are still concerns about the long-term regulation of AI in the European Union.⁹

⁸ At this time, there was significant news coverage indicating that the ban would restrict access to ChatGPT. There were no enforcement actions against OpenAI because it immediately complied by banning all IP addresses from Italy. It is also clear that some users may have accessed ChatGPT via VPNs (a foreign IP address). However, while VPNs might mute the effects, we think that VPNs are unlikely to completely nullify the ban, since not everyone is familiar with VPNs. Furthermore, VPNs can be slow and tend to absorb all web traffic. In addition, some corporate networks may not allow VPNs for reasons of network security and monitoring.

⁹ EU regulators have been revising laws to regulate AI (“EU lawmakers vote for tougher AI rules as draft moves to the final stage,” Reuters, June 14, 2023).

2.2. Conceptual Framework

2.2.1 Information Production by Financial Analysts

An extensive amount of prior literature relies on noisy rational expectations theories (Grossman and Stiglitz, 1980; Verrecchia, 1982) to point out limits to the efficiency of capital markets. When acquiring information is costly, market prices cannot fully reflect the knowledge of well-informed investors, because investors must make trading profits to cover their information acquisition costs. Sims (2003, 2006) and Veldkamp (2006) have explored how individuals and institutions, constrained by their capacity to process information, fail to analyze publicly available information, resulting in systematic noise that can skew or delay information incorporation into prices. The theory explains why, when information production is costly, rational investors tend to access only a subset of available information, thereby affecting the informational efficiency of prices. This perspective has inspired a growing body of empirical studies on the economic consequences of information processing limitations, as viewed through the lens of rational inattention (e.g., Blankespoor et al., 2014; Dong et al., 2016; Blankespoor et al., 2020).

In this vein, Blankespoor et al. (2020) suggest that investing in technology may help mitigate information processing constraints. However, the costs typically associated with technology adoption imply that only a select group of agents reaps the benefits (Blankespoor et al., 2014). The introduction of generative AI, led by OpenAI's ChatGPT, may have altered these dynamics. Known for its contextual awareness and coherent conversations, ChatGPT quickly gained popularity due to its minimal adoption costs, user-friendly interface, and integration into popular platforms and apps.

How does the adoption of ChatGPT affect information processing? We rely on the ChatGPT ban in Italy, which disrupts the usage of the technology, to assess how an increase in processing

costs affects information flows in the capital market. We posit that the ban's impact may have been most evident in the behavior of financial analysts. First, analysts play a well-documented role in facilitating information flow within the capital market. They process macro- and micro-level data, distilling complex information into reports that provide value-relevant insights. These reports aid both sophisticated and less sophisticated investors in making informed decisions. Second, although ChatGPT is a widely used application, its adoption has been uneven and often depends on users' technological proficiency. Therefore, we assume that financial analysts with strong technical backgrounds would be more likely to embrace such innovative tools.

During the ban, domestic financial analysts faced greater information processing constraints than foreign analysts. This restriction likely decreased their ability to process information quickly and effectively, leading to reduced or lower-quality information production. ChatGPT can assist financial analysts by streamlining routine, non-research tasks, thereby freeing up more time for critical activities, such as conducting in-depth firm-specific research. Furthermore, ChatGPT potentially aids in financial statement analysis and earnings predictions (Chen et al., 2024; Kim et al., 2024). Without the aid of ChatGPT, analysts may incur higher costs in processing vast amounts of information. Furthermore, analysts with limited time and resources may prioritize easier-to-process industry-level information, which may result in important firm-specific insights being overlooked (e.g., Peng, 2005; Peng and Xiong, 2006; Blankespoor et al., 2020).

2.2.2 Informational Efficiency

Given the important role of analyst forecasts in enhancing informational efficiency (Womack, 1996; Healy and Palepu, 2001; Bradshaw, 2011; Bradshaw et al., 2016), the ban may have reduced informational efficiency by decreasing the frequency and quality of analyst forecasts. If the ban limited the provision of forward-looking information by analysts, less information would be

integrated into stock prices ahead of events. In markets where traders rely on analyst forecasts to inform their trading strategies, the absence of such information suggests that fewer market participants are informed before earnings announcements. Since private information is gradually incorporated into prices through trading (Kyle, 1985; Back, 1992), the lack of information flow from analysts could result in a more pronounced price reaction to earnings announcements.

The reduction in both the quantity and quality of analyst forecasts due to the ban may also have affected bid–ask spreads (Glosten and Milgrom, 1985). *A priori*, the impact of increased processing costs on bid–ask spreads hinges on whether the ban affected traders with less information, which would be consistent with theories of information leveling the playing field, or predominantly affected traders with more information, which would align with theories of information complementarity.¹⁰

In the context of studying the practices of financial analysts, prior empirical studies generally support the “leveling the playing field” and show that analyst information production helps to reduce information asymmetry (Mikhail et al., 2007; Kelly and Ljungqvist, 2012; McLean et al., 2024). This is because analyst reports are widely accessible and provide readily usable analyses. Hence, reduced analyst information intermediation may increase information asymmetry among different groups of investors. While both large and small investors rely on analyst reports for informed trading, larger and more sophisticated investors may be less affected by the ban due to their access to extensive resources. In contrast, smaller and less sophisticated investors might lack the means to compensate for reduced analyst information production, leaving them relatively less

¹⁰ On the one hand, enhanced information processing may reduce information asymmetry, which would potentially result in the leveling of the playing field among investors. On the other hand, if not all investors process information equally, it could increase information asymmetry by disproportionately assisting more sophisticated investors in collecting and integrating additional or private information (Kim and Verrecchia, 1994; Blankespoor et al., 2014; Cheynel and Levine, 2020; Aghamolla and Smith, 2023).

informed. This disparity could intensify information asymmetry in the market, leading to wider bid–ask spreads.

III. The Impact of the ChatGPT Ban on Analyst Information Production

This section investigates whether the ban affects financial analysts’ information production. We perform multiple tests on the impact of the ban on analysts. First, we employ commonly used techniques from computer science to detect AI-generated content in the analyst reports. This test helps address two questions: 1) whether the analysts use generative AI in drafting their reports, and 2) whether the usage of generative AI in the analyst reports is disrupted during the ban. Second, we assess the impact of the ban on domestic analysts’ information production, including forecast frequency and accuracy. Finally, we test whether foreign analysts take up the slack, given the potential disruptions in information production.

3.1. Sample and Data Sources

We employ three datasets. First, we construct an analyst-level dataset that captures each analyst’s information production through analyst reports. Second, we build a dataset at the firm–analyst level, which measures analyst forecast frequency and accuracy for each public firm. Third, we compile a firm-level dataset comprising firm-level outcomes, such as bid–ask spreads and trading volumes. Collectively, these datasets allow us to assess the impact of the ban on analysts’ information production and the associated capital market consequences.

To build the first dataset, we obtain analyst reports on listed firms in Italy through Thomson One’s Investtext. For each analyst report, Investtext provides details on the Document Date (when the report was written), Contributor (the brokerage firm’s name), Author, Language, and Pages. In addition, we extract firm names and tickers, as well as analyst email addresses, physical addresses,

and phone numbers. We match the analysts' reports with the firms' information from Refinitiv Eikon using firm names and tickers.

To identify domestic (i.e., Italy-based) analysts, we use office addresses, email domains, or phone numbers.¹¹ Our final sample comprises 2,090 reports authored by 677 analysts, 169 of whom were based in Italy.¹² We manually search each analyst in our sample on LinkedIn using their names and the names of their affiliated brokerage houses. We gather data on analysts' educational backgrounds, work experience, and other characteristics. After matching data on analyst reports with LinkedIn profiles, we obtain a sample of 1,208 analyst-month observations involving 602 analysts affiliated with 54 brokerage houses.¹³

We construct the second dataset to examine analyst forecast frequency and accuracy. We obtain data on analyst forecasts with a one-year forecast horizon from I/B/E/S.¹⁴ After matching the I/B/E/S forecast data with LinkedIn data on analyst characteristics, we compile a sample of 2,084 analyst-firm-month observations, involving 155 analysts issuing forecasts for 107 firms, of whom 49 were domestic analysts. Table 1 provides the summary statistics for the analyst-firm-level sample of analyst forecasts. On average, each analyst issued 0.33 forecasts per month for Italian firms, and 30% of analysts issued at least one forecast monthly. The analysts in this sample

¹¹ Analysts are classified as domestic if their office address was located in Italy. When the address was unavailable, we determined their classification by checking whether their email domain ended with ".it" (the country code for Italy) or if their office phone number began with Italy's country code "+39". We manually verified this information to ensure accuracy.

¹² In our sample, approximately 98% of the analyst reports were written in English, while the remaining 2% were in Italian.

¹³ Table OA1 in the online appendix presents the covariate balances between domestic and foreign analysts' reports during the pre-ban period from January 1 to March 30, 2023. Across all analyst characteristics, forecast frequency, and forecast quality measures, there were no statistically significant differences between domestic and foreign analysts at the 5% level, indicating that the analysts were comparable and exhibited similar forecasting behavior prior to the ban. After matching, Italian firms and their foreign counterparts have insignificant differences in market capitalization and ROA.

¹⁴ Each forecast was associated with a Marked Analyst Code. We then retrieved the analyst's surname and the initial of their given name from the Analyst Recommendation File and matched this information to the forecast data using the Marked Analyst Code.

had an average of 11.4 years of experience, and 50.5% held technical degrees.

To construct the third firm-level dataset, we obtain stock returns, trading volumes, and ownership data from the Refinitiv Eikon and Datastream databases. This dataset includes firms from the European Union and the United Kingdom. We employ propensity score matching to create a group of control firms to alleviate concerns about the differences between Italian and non-Italian firms. We match each Italian firm to a foreign firm based on industry (classified by a two-digit NAICS code), market capitalization, and profitability (measured by return on assets (ROA)) without replacement. Market capitalization is calculated as of March 24, 2023, one week before the ban. For our firm-level analysis, we use a control sample of firms in the European Union (excluding Italy) and the United Kingdom, resulting in a sample of 228 Italian firms and 228 matched European firms.¹⁵ Table 1 presents the summary statistics. To mitigate the influence of outliers, we winsorize all continuous variables at the first and 99th percentiles, excluding stock returns and unexpected earnings. The median firm in our sample has a market capitalization of EUR 315 million and an ROA of 0.05.

We define the ban period as March 31 (inclusive) to April 28 (inclusive), totaling 21 working days. For comparability, each monthly time unit in the pre-, during-, and post-ban periods included exactly 21 working days. The pre-ban period consists of three consecutive subperiods with 21 working days (January 2 to March 30), referred to as January, February, and March for simplicity. The post-ban period similarly consists of two consecutive 21-working-day subperiods (April 29 to June 27), referred to as May and June.

¹⁵ It is conceivable that investors may have anticipated a ban in other European countries, as evidenced by announcements of investigations in France (April 11), Spain (April 13), and Germany (April 24), thereby muting the documented effect if control firms were also potentially treated. To address this concern, we show in an unreported table that our results are similar to those using a matched control sample with only UK firms.

3.2. Evidence of the Likelihood of AI-Generated Content in Analyst Reports

We investigate whether the ban affects analysts’ utilization of AI in writing their reports. We use two established metrics from the computer science literature that help with AI-generated text detection (AGTD): perplexity and burstiness. Perplexity and burstiness are broadly used in the natural language processing (NLP) literature as benchmark measures for AGTD (Mitchell et al., 2023; Li et al., 2023; Wang et al., 2023; Chakraborty et al., 2023).¹⁶ A text displaying higher levels of perplexity and burstiness is more likely to be authored by a human.

One caveat is that more recent large language models (LLMs) have demonstrated the utilization of a wide range of vocabulary (OpenAI, 2023), which may potentially lower the detection power of perplexity- and burstiness-based AGTD algorithms. We conduct a simple experiment to alleviate this concern.¹⁷ We find that perplexity and burstiness scores are significantly higher in human-only written reports compared to AI-polished and AI-only written reports. Specifically, perplexity and burstiness decrease by 14.2% and 25.5%, respectively, in AI-polished reports and by 16.7% and 35.0% in AI-only written reports, which supports the

¹⁶ Perplexity indicates the procedural nature of AI text generation, noting that humans exhibit significant variation in linguistic constraints, vocabulary, syntax, and other factors, while LLMs display a higher degree of consistency in their linguistic style and structure. Burstiness measures how much perplexity varies over the entire document on the grounds that AI-generated text displays a higher frequency of clusters of similar words or phrases within text sections. In contrast, humans exhibit a broader variation in their lexical choices, with a more extensive range of vocabulary being used.

¹⁷ This study has been approved by the Institutional Review Board for Human Participants (IRB) as required by our universities. In this experiment, we used perplexity and burstiness measures to compare texts written by humans versus texts written with different degrees of ChatGPT 3.5 involvement. Specifically, we ask 35 participants to summarize the earnings call transcript on the 2022 results of Prada S.p.A. (PRDSY). The participants are from a major research university with either accounting or finance backgrounds. The majority of participants in our experiment are not native English speakers, which is consistent with the sample of Italian analysts in our empirical analyses. Given that many analysts based in Italy are also unlikely to be native English speakers, they may have a greater need for ChatGPT compared to native English speakers. Starting with human-only written reports, we use ChatGPT 3.5 to generate two additional versions: AI-polished reports, where the human-written content was refined by the AI, and AI-only written reports, which was created by prompting ChatGPT 3.5 to summarize raw earnings call transcripts. We then assess the perplexity and burstiness scores across these three versions: human-only written, AI polished, and entirely AI generated (see Appendix B for details).

effectiveness of AGTD measures in detecting AI-generated content.

Next, we utilize AGTD measures to identify ChatGPT’s footprint on analyst reports. In particular, we compute the perplexity and burstiness levels for each analyst report developed by professional analysts. Then, we create a composite score of perplexity and burstiness (*Composite Score*) by averaging the scores of the two measures. We aggregate the data into a panel at the analyst-month level to investigate how analysts’ utilization of AI changes from the pre-ban period to the ban period. Our final sample comprises 1,208 analyst-month observations, including 893 observations from the pre-ban months and 315 observations during the ban.

We then employ a difference-in-differences design to estimate whether the prominence of human writing increases for domestic analysts relative to foreign analysts:

$$Text\ Characteristics_{a,t} = \beta_1 Italy\ Analyst_a \times Ban_t + \beta_2 Italy\ Analyst_a + Controls + \alpha_t + \gamma_n + \epsilon_{a,t}, \quad (1)$$

where $Text\ Characteristics_{a,t}$ is proxied by the perplexity, burstiness, and composite score of perplexity and burstiness (*Composite Score*) for the reports by analyst a in month t . As explained earlier, heightened perplexity and burstiness indicate a greater likelihood of human writing. *Italy Analyst* is a dummy variable indicating domestic analysts, and *Ban* is a time dummy that equals one for the ban period and zero for the pre-periods from January to March 2023. Control variables include a set of analysts’ characteristics manually collected from LinkedIn, including work experience, social connection, and the highest degree obtained. We add brokerage-house fixed effects (i.e., γ_n) to account for the effect of idiosyncratic brokerage-house-level policies on ChatGPT usage. We do not include analyst fixed effects due to data limitations. Although the Investext database remains one of the most comprehensive sources for international analyst reports, it covers only a subset of actual analyst forecasts. Incorporating analyst fixed effects would require data points for each analyst both before and after the ban (e.g., Breuer and deHaan, 2024). Despite

this constraint, our results (untabulated) remain consistent with the inclusion of analyst fixed effects, although the statistical significance is diminished. We add month fixed effects (i.e., α_t) to absorb time trends in ChatGPT adoptions. Standard errors are clustered at the analyst level.

Table 2, Columns 1 to 3, present our results. The likelihood of domestic analysts' reports being written by humans significantly increased during the ban across all three proxies compared to foreign analysts. Regarding economic magnitudes, perplexity, burstiness, and the aggregate measure increased by 39.5%, 61.6%, and 69.5%, respectively, of the sample standard deviation for domestic analysts. Overall, our findings provide the first piece of suggestive evidence that analysts used ChatGPT when preparing reports, and that the unexpected ban affected their output.

We then conduct a cross-sectional test based on which subgroups of analysts are more likely to adopt ChatGPT before the ban and, thus, should be more affected by the ban. Analysts with a more technical education background are potentially more inclined to adopt ChatGPT and employ it to write their reports. Empirically, we construct a partitioning variable based on the analysts' educational background. We follow Frank et al. (2023) in defining analysts as possessing technical skills if they majored in technical subjects during their undergraduate or postgraduate studies (*Tech Major*).¹⁸ If some analysts adopted ChatGPT, as suggested by Table 2, Columns 1 to 3, we should observe a more pronounced effect among analysts predisposed to experiment with new technologies, such as those with technical degrees.

¹⁸ These majors are mostly STEM majors. See Table A1 in Frank et al. (2023) for a detailed list of technical majors. We caveat that educational backgrounds remain a noisy measure of technology adoption. It is implausible that STEM skills could be applied when using ChatGPT (e.g., writing an API or code), but we conjecture that educational choices and prior exposure to science and technology may be associated with a greater propensity to experiment with new technologies. According to Bick et al. (2024), the US adoption of generative AI by STEM majors has outpaced the adoption of generative AI by other majors. For example, data from the June 2024 wave of the Real-Time Population Survey reveals that 44.8 percent of workers with STEM degrees use generative AI in their jobs, compared to 29.5 percent of workers with degrees in business, economics, or communication, and 21.6 percent of those with degrees in other fields, such as liberal arts and humanities.

Table 2, Columns 4 to 6, present our findings. The three-way interaction between *Italy Analyst*, *Ban*, and *Tech Major* is significantly positive for perplexity and the aggregate measure. Perplexity, burstiness, and the aggregate measure increased by 60.6%, 37.5% (although statistically insignificant), and 66.7%, respectively, of the sample standard deviation for domestic analysts majoring in technical degrees relative to their peers during the ban. Furthermore, the two-way interaction terms between *Italy Analyst* and *Ban* are generally positive for all three measures, suggesting that the ban negatively affects even domestic analysts not majoring in tech-related degrees. Overall, our results imply that the increase in the probability of using human writing is stronger in the subgroup of analysts who are potentially more intensive users of ChatGPT in the pre-ban periods.

3.3. *Impact of the ChatGPT Ban on Analysts' Forecast Frequency*

Having documented preliminary evidence of analysts' use of ChatGPT, we explore ChatGPT's impact on information production. The empirical challenge is that the ban may have influenced both analysts' information processing capabilities and capital market participants' expectations of fundamental firm changes. As a result, observed changes in analysts' behavior could have stemmed from changes in information processing costs or firm fundamentals.

To address this challenge, we employ an empirical strategy that involves conducting comparisons within the set of analysts covering the same Italian firm. This research design enables controlling for variations in firm fundamentals. Consequently, any documented effect should have come from the ban's impact on analysts' information processing capacities. Our firm-analyst-month-level dataset consists of 2,084 observations, with 1,563 observations from the pre-ban periods and 521 observations from the ban period. Specifically, we assess the difference in the

changes in forecast frequency between domestic versus foreign analysts using the following specification:¹⁹

$$Forecast\ Frequency_{i,a,t} = \beta_1 Italy\ Analyst_a \times Ban_t + \alpha_{i,t} + \gamma_{i,a} + \epsilon_{i,a,t} \quad (2)$$

where $Forecast\ Frequency_{i,a,t}$ is the number of forecasts issued by analyst a on firm i in month t .

Table 3 presents our results. We observe a significant decrease in the number of forecasts by domestic analysts compared to foreign analysts during the ban. The results are robust when using multiple specifications with different sets of fixed effects. Column 1 shows the results after controlling for firm, analyst, and month fixed effects. Column 2 (3) shows the findings after controlling for a tighter set of fixed effects of analyst and firm-by-month (firm-by-month (i.e., $\alpha_{i,t}$) and analysts by firm (i.e., $\gamma_{i,a}$)) fixed effects. These high-dimensional fixed effects absorb the control variables.

Regarding economic magnitudes, the number of forecasts decreased by around 0.11 (21% of the sample standard deviation) for domestic analysts relative to their foreign peers. Our findings are consistent with the idea that ChatGPT assists analysts in their day-to-day operations. However, we are ambivalent about how analysts actually use ChatGPT, because we only observe the occurrence of a forecast. ChatGPT could help analysts process information or assist them in better explaining and presenting their forecasts. Both channels are consistent with a reduction in the information available to investors.

We conduct cross-sectional tests on the ban's impact on different groups of analysts. Analysts more likely to have adopted ChatGPT before the ban are expected to have been more affected. We employ two proxies to measure the likelihood of ChatGPT usage. First, we construct a dummy

¹⁹ We find similar results using the Poisson regressions.

variable, *Ex-Ante Prob (AI-Generated Text)*, which is set to one if the analyst’s reports in the pre-ban period were more likely to be written by AI (i.e., the composite score of perplexity and burstiness (*Composite Score*) was lower than the sample median). Second, we use *Tech Major*, which is based on whether analysts majored in technical subjects. Table 4 presents our findings. The decline in information intermediation during the ban was concentrated among domestic analysts with a technical degree or with a higher chance of using ChatGPT to prepare reports in the pre-ban period.²⁰

3.4. Do Foreign Analysts Take Up the Slack?

Given analysts’ reduced information production, a crucial empirical question is whether foreign analysts compensate to fill the gap in the information environment. A substitution effect would imply that the provision of information by domestic (foreign) analysts decreases (increases) during the ban.²¹ To examine this question empirically, we revisit the provision of information by domestic and foreign analysts during the ban, which is shown in Table 5. We estimate the effect of the ban on the number of forecasts by domestic analysts in Column 2, the corresponding effect on foreign analysts in Column 5, and the combined amount of forecasts by domestic and foreign analysts in Column 1. Columns 3 and 4 (6 and 7) further partition the domestic (foreign) analysts by the ex-ante probabilities of AI-generated content in their reports before the ban and their educational backgrounds (i.e., technical analysts).

²⁰ In the online appendix, we show the results of our two additional robustness checks. First, as shown in Table OA2, we obtain consistent results using a dummy variable (*Forecast Dummy*) to indicate whether an analyst issued at least one forecast during a month, which allowed us to characterize the reduction in analyst output through extensive margins. Second, we conduct placebo tests, as shown in Table OA3, to verify that analyst characteristics other than *Tech Major* did not drive our results, including the number of connections and work experience.

²¹ Such substitution may have biased the causal effect by violating the stable unit treatment value assumption (SUTVA), as foreign analysts were treated by the change in information and increased their information provision. In particular, while the presence of this substitution could not have occurred absent a decrease in information provision by domestic analysts, the estimated effect may have been biased upwards if the control sample endogenously increased their information provision.

We note several findings. First, we find a significant decrease in the number of forecasts by domestic analysts (Column 2), and the effect was stronger for analysts with a higher propensity to use ChatGPT (Columns 3 and 4). Second, we do not find any significant changes in information production by foreign analysts from the period before the ban to during the ban (Column 5), and there are no significant differential impacts across foreign analysts with different propensities for using ChatGPT (Columns 6 and 7). Finally, turning to the total number of forecasts by domestic and foreign analysts for Italian firms in Column 1, our evidence suggests that net information production was likely to have decreased, though the coefficient is statistically insignificant, potentially due to noise and testing power. Overall, unless there was an unrelated decrease in the propensity to issue forecasts by foreign analysts during the ban, our evidence is inconsistent with foreign analysts substituting for the reduced information provision by domestic analysts.

3.5. Impact of the ChatGPT Ban on Analysts' Forecast Accuracy

In the previous sections, we demonstrated that the ban reduces the quantity of forecasts produced by domestic analysts. However, informational efficiency depends not only on the quantity of information, but also on its quality. Although we present evidence in Table 2, suggesting that the analysts may have used ChatGPT to assist in drafting reports, greater use need not translate into higher-quality information provision, since the latter requires analysts to have the ability to productively employ AI to search and process value-relevant information.

We rely on forecast accuracy as a proxy for information intermediation quality. If the analysts had reached an advanced level of utilization, the ban may have adversely impacted this capability, thereby potentially reducing the precision of their forecasts. Therefore, we assess how forecast accuracy changes from before to during the ban for domestic versus foreign analysts using the following specifications:

$$Forecast\ Accuracy_{i,a,t} = \beta_1 Italy\ Analyst_a \times Ban_t + Controls + \alpha_t + \gamma_{i,a} + \epsilon_{i,a,t}, \quad (3)$$

where $Forecast\ Accuracy_{i,a,t}$ is the absolute forecast error for analyst a covering firm i in month t , calculated as the absolute value of the difference between the actual and forecasted values of earnings per share deflated by the stock price two days before the earnings forecasts:

$$AFE_{i,a,t} = \left| \frac{Analyst\ Forecasted\ EPS_{i,a,t} - Real\ EPS_{i,t}}{P_{i,t-2}} \right|. \quad (4)$$

We control for a vector of lagged firm-level characteristics, which consists of the book-to-market ratio, idiosyncratic volatility, market capitalization, and stock return over the last month. We further include analyst-by-firm (i.e., $\gamma_{i,a}$) and month fixed effects (i.e., α_t).²² Standard errors are clustered by industry.

Table 6 reports the effects of the ban on analyst forecast accuracy. Column 1 shows a significant increase of 0.95% (8% of the sample standard deviation) in the absolute forecast error by domestic analysts compared to their foreign peers during the ban. We further interact the $Italy \times Ban$ term with two proxies for usage: *Ex-Ante Prob (AI-Generated Text)* and *Tech Major* in Columns 2 and 3, respectively. We find that the decline in forecast accuracy during the ban was more pronounced for domestic analysts with a technical degree or higher ex-ante utilization in the pre-ban period. Moreover, the negative coefficient of the two-way interaction term $Ban \times D$ suggests that AI-savvy foreign analysts improved their forecasting in the post-ban period, which could be due to the broader trend of increased use of ChatGPT and related technological tools over time. Collectively, our evidence indicates that the ban potentially diminished analysts' capacity

²² We do not control for firm-by-month fixed effects due to sample size limitations. We cannot compute forecast accuracy for firms that do not have any analyst forecasts, which reduces the sample size and, consequently, the within-firm-month variation. To alleviate the concern that our results were driven by time-varying differences across firms, we include firm-by-analyst and month fixed effects and a vector of time-varying firm-level characteristics, as shown in Table 6.

for information processing and impaired their ability to provide accurate information to capital market participants.

3.6. Industry/Macro-Level Information in Analyst Reports

Another approach to assessing the impact of the ban on the quality of analysts' information intermediation is to analyze the specific types of information the analysts discussed in their reports. This approach also helps to illuminate how the analysts utilized ChatGPT during the information processing stage, and how effectively it could be integrated into the analytical process. Thus, we examine the extent to which ChatGPT aided the analysts in processing industry- or macro-level information. We focus on this type of information because it is plausibly relatively less idiosyncratic, easier to process, and more frequently utilized when information acquisition is costly (Peng, 2005; Peng and Xiong, 2006).

Following Li et al. (2021), to identify the fraction of industry information discussed in analyst reports, we rely on the frequency of industry-related words and construct a variable named *Industry Information*. We select “industry,” “macroeconomy,” and “market” as seed words and employ the word embedding model, “word2vec,” to learn all similar words in the analyst reports. Then, we select a comprehensive industry-related word list based on the output of the word embedding model (see Appendix C). Based on this list, we define *Industry Information* for all reports issued by analyst a in month t as

$$Industry\ Information_{a,t} = \sum_{r=1}^{N_{a,t}} \frac{Industry\ Words_{r,a,t}}{Length_{r,a,t}} / N_{a,t},$$

where $Industry\ Words_{r,a,t}$ is the number of all industry-related words in report r issued by analyst a in month t , and $Length_{r,a,t}$ is the number of words in report r by analyst a in month t . $N_{a,t}$ is the

number of reports issued by analyst a in month t . We multiply *Industry Information* by 100 to express it as a percentage. We follow the regression specification in Equation (1).

Table 7, Column 1, indicates a significant increase of 8.43% (relative to the sample mean) in *Industry Information* for domestic analysts compared to their foreign peers during the ban. The effect is stronger for domestic analysts with higher ex-ante ChatGPT utilization in the pre-ban period or a technical degree (Columns 2 and 3).²³ The negative coefficient of the $Ban \times D$ interaction term suggests that AI-savvy foreign analysts incorporated relatively more firm-specific information (or less industry-related information) in the ban period. This provides suggestive evidence of the enhanced information-gathering capacities facilitated by ChatGPT over time. Together with the findings on forecast accuracy, our interpretation is that the ban discouraged analysts from effectively processing firm-specific information, leading them to rely more on industry- and macro-level information. This shift could have contributed to lower forecast accuracy if industry-level information was coarser than firm-specific information.

IV. The ChatGPT Ban and the Information Environment

This section investigates the ban's effect on the information environment. First, we study the market response to earnings announcements during the ban to shed light on investors' overall reaction to public information. Second, we examine bid–ask spreads to assess whether the ban harms investors' information processing capabilities (especially smaller investors) and increases information asymmetry.

²³ Our findings do not necessarily imply that analysts used ChatGPT to process only industry-specific information. Instead, consistent with Peng and Xiong (2006), they may, at the margin, have re-allocated more attention to more easily processed industry and macro-level information than firm-specific information once their processing capacity was reduced.

4.1. Impact of the Ban on Market Reactions to Earnings Announcements

As discussed above, if the ban limits analysts' provisions of forward-looking information, less of this information would be integrated into stock prices before the earnings announcements. Hence, the market reaction to earnings announcements would likely be stronger, as earnings news is largely new to the market (Ball and Shivakumar, 2008).²⁴

Following Livnat and Mendenhall (2006) and Mian and Sankaraguruswamy (2012), we calculate the unexpected earnings (UE), which represent the unexpected part of the earnings announcement, as follows:

$$UE_{i,t} = \frac{Earnings_{i,t} - Earnings_{i,t-1}}{P_{i,t}}, \quad (5)$$

where $Earnings_{i,t}$ is the EPS of firm i in year t , and $P_{i,t}$ is the stock price two days before the earnings announcement date in year t (to avoid a contamination of announcements on prices). We use the prior year's earnings ($Earnings_{i,t-1}$) as investors' expected earnings for year t . For robustness, we follow Chiang et al. (2019) in calculating robust earnings surprises and find that our results are qualitatively unaffected.²⁵

We then measured the stock market's reaction to earnings announcements by regressing stock returns on unexpected earnings:

²⁴ Note that the opposite finding is also possible. If the ban restricted investors' information processing capacity, they may have struggled to digest earnings information efficiently during the ban period. This reduced capacity could have led to delayed or less accurate assessments of earnings reports, resulting in a delayed market response.

²⁵ We note that unexpected earnings are known to have significant outliers (e.g., Collins and Kothari, 1989; Kothari, 2001; Gipper et al., 2020). Hence, we truncate UE at the 5% and 95% levels to mitigate the outliers' effect. Our results would be similar if we truncated unexpected earnings at the 1% and 99% levels. In untabulated analyses, we follow Chiang et al. (2019) in calculating a parametric-free measure of earnings surprises. We calculate the fraction of misses on the same side (FOM). Intuitively, supposing that there are N analyst forecasts, K would be the number of forecasts less than the actual announced earnings (A), and M would be the number of forecasts greater than A . The fraction of misses (FOM) on the same side would be given by $K/N - M/N$, which would take on values between -1 and 1. The higher the FOM, the more positive the earnings surprise. Our results remain qualitatively similar.

$$Ret_{i,t} = \beta_1 Italy_i \times Ban_t \times UE_{i,t} + \beta_2 Italy_i \times Ban_t + \beta_3 Italy_i \times UE_{i,t} + \beta_4 Ban_t \times UE_{i,t} + \beta_5 UE_{i,t} + \beta_6 Italy_i + \beta_7 Controls \times UE_{i,t} + Controls + \alpha_t + \gamma_j + \epsilon_{i,t}, \quad (6)$$

where $Ret_{i,t}$ is the cumulative abnormal return for firm i in four different time windows: one, three, five, or seven days centered around the earnings announcement dates.²⁶ Following Gipper et al. (2020), we included a set of firm-level controls: *Loss*, *Size*, *Market-to-book*, *Leverage*, *Earnings Persistence*, and *Beta*. We further controlled for industry (i.e., γ_j) and month fixed effects (i.e., α_t), as well as the interaction between the control variables and *UE*, and clustered standard errors by industry. We are unable to include firm fixed effects due to the small number of observations (192).

Table 8 shows that the market reaction to earnings announcements is stronger for domestic firms than for comparable foreign firms during the ban period compared to the pre-ban period. With a one-standard-deviation (0.045) increase in unexpected earnings, Italian firms experience an additional positive market-adjusted return of 3.73 percentage points during the ban relative to their peers on the earnings announcement day. The additional cumulative abnormal return is 2.51, 2.75, and 3.72 percentage points for the three-day, five-day, and seven-day windows around the announcement dates, respectively.

We shift our focus to other coefficients in Table 8. The main effect (*UE*) is positive and statistically significant. The economic magnitude is also broadly consistent with the findings in the prior literature. The estimated coefficient of $Ban \times UE$ is significantly negative, which captures

²⁶ We take the difference between daily stock and market returns as daily abnormal returns. Our results are robust to using an alternative measure of abnormal returns adjusted based on the CAPM model. To alleviate endogeneity concerns, we employ propensity score matching to construct a set of control firms. Each Italian stock is matched to a non-Italian stock based on industry (classified by a two-digit NAICS code), market capitalization, and profitability measured by return on assets (ROA) without replacement. The market capitalization for each firm is calculated as of March 24, 2023 (i.e., one week before the ChatGPT ban). After matching, Italian and matched firms are found to have had insignificant differences across market capitalization and ROA.

the change in market responsiveness to unexpected earnings for non-Italian firms between the pre- and during-ban periods. The negative coefficient suggests that earnings contain less news content over time (i.e., when comparing the ban period with earlier periods) in countries with ChatGPT due to the faster incorporation of the firms' information into prices before the earnings announcement. The coefficient of $Italy \times UE$ is negative and significant, implying that, relative to other countries, domestic firms experience weaker market reactions to earnings announcements in the pre-ban period. Finally, the coefficient of $Italy \times Ban$ is significantly positive, indicating that domestic firms experienced more positive returns during the earnings announcement windows compared with firms from foreign countries in the pre-ban period. Overall, our results suggest that the ban negatively affects investors' information processing prior to earnings announcements.

4.2. Impact of the ChatGPT Ban on Bid–Ask Spreads

Next, we examine the impact of the ChatGPT ban on bid–ask spreads by using a difference-in-differences design:

$$Bid\text{-}ask_{i,t} = \beta_1 Italy_i \times Ban_t + Controls + \alpha_i + \gamma_t + \epsilon_{i,t}. \quad (7)$$

We focus on the bid–ask spreads and the bid–ask spreads adjusted for the trading volume. We control for firm and month fixed effects. Standard errors are clustered at the industry level. Table 9 shows that both the raw (Column 1) and adjusted (Column 2) bid–ask spreads increased significantly by about 0.03, which is about 1.5 times the sample standard deviation.²⁷ This result is consistent with the ban hindering information processing and increasing information asymmetry across investors.

²⁷ We show in Table OA4 similar results on total volatility of stock return, idiosyncratic volatility, and turnover. Turnover declined significantly during the ban for Italian stocks.

We also conduct several cross-sectional tests on the adjusted bid–ask spreads (Table 10) to shed light on the mechanisms driving the increased information asymmetry. Firms with limited analyst coverage (Column 1) are more prone to information processing constraints and experience decreased liquidity during the ban. Firms with higher foreign ownership (Column 2) exhibit a smaller increase in information asymmetry, because the ban applies only to Italian geographic boundaries and does not directly affect foreign investors. Consistent with smaller and unsophisticated investors relying more on technology to parse through financial information, firms with higher institutional ownership (High Institution, Column 3) tend to exhibit a smaller increase in bid–ask spreads. Overall, our results point to the role of generative AI as a productivity tool that facilitates investors’ information processing.

V. Additional Analyses

5.1. Dynamic Treatment Effects of the ChatGPT Ban

We estimated the dynamic treatment effects of the ban on major outcomes from periods before the ban to periods after the restoration. Our tests serve two main purposes. First, we assess the parallel trend assumptions by examining the pre-trends for treated and control analysts or firms in the periods leading up to the ChatGPT ban. Second, examining the periods after the restoration of ChatGPT allows us to assess whether the treatment effect reverses after the ban is lifted.

We extend our sample to June 2023 and estimate the dynamic treatment effects for three sets of major outcomes. First, we examine the three different measures of the probability of AI-generated content in analyst reports. Empirically, we estimate the following specification:

$$Text\ Char_{a,t} = \beta_s \sum_{s=-3 \sim +2} Italy\ Analyst_a \times D_{s(a,t)} + \beta_3 Italy\ Analyst_a + Controls + \alpha_t + \gamma_n + \epsilon_{a,t}, \quad (8)$$

where $Text\ Char_{a,t}$ is the score of AI-generated texts proxied by *perplexity*, *burstiness*, and a

composite score of perplexity and burstiness (*Composite Score*) for analyst a in month t . $D_{s(i,t)}$ is a set of indicator variables that equals one if the time period is s months away from April 2023. Our sample period ranges from three months before the ban (i.e., January 2023) to two months after the ban (i.e., June 2023), with January 2023 ($s = -3$) omitted and used as the benchmark period.

Figure 1 presents our results. We find no significant disparities in AI-generated content in the analysts' reports in the periods leading up to the ban, which is consistent with the parallel trends assumption. There is a spike in the probability of human-generated content for domestic analysts during the ban (due to the ChatGPT ban) compared to their international counterparts. After ChatGPT was restored, the differences between domestic and foreign analysts' AI-generated contents disappeared in the two months post-periods.

Second, given the evidence on AI-generated content in analyst reports, we study the dynamic treatment effect on analysts' forecast frequency. We plot the estimated coefficients in Figure 2. We find that domestic analysts decreased their forecast frequency during the ban period ($s = 0$), and the effect reverses in the two months after ChatGPT was restored in Italy ($s = +1$ to $+2$). We note that the estimated coefficient for the first month after the restoration (i.e., $s = +1$) is positive but insignificant due to large standard errors. One possible explanation is that after ChatGPT was restored, the re-adoption among analysts is unbalanced, with some analysts being slow to return to prior routines due to the disruptions of the ban. This heterogeneity leads to greater standard errors in the estimates, making the coefficients statistically insignificant shortly after the reactivation. The reversal is statistically significant in the second month after the ban was lifted ($s = +2$).

In addition to the effects on analyst information production, we plot the dynamic impact of the ban on information asymmetry among investors (i.e., bid–ask spreads) in Figure 3. We find no

significant disparities in bid–ask spreads (adjusted bid–ask spreads) across domestic and foreign firms during the periods leading up to the ban in Panel A (Panel B). There is a spike in the bid–ask spreads during the ban, consistent with greater information asymmetry among investors. After ChatGPT is restored, the differences in bid–ask spreads between Italy and matched control firms reverse back to zero over the subsequent two months.

VI. Conclusion

In this study, we examine the effect of a major technological change on investor information processing by examining the ChatGPT ban in Italy in April 2023 and developing new insights into whether changes in processing affect information flows and asymmetric information. We demonstrate that the ban reduces the likelihood of usage by domestic analysts, as suggested by an increase in the perplexity and burstiness of their reports. The increase in perplexity and burstiness is more pronounced among analysts with technological backgrounds. Then, we show that a reduction in usage discourages domestic analysts from issuing forecasts compared with foreign analysts covering the same firm. The reports written by domestic analysts also exhibit declines in forecast accuracy and a greater reliance on industry- and macro-level information provision. These negative effects are more pronounced for analysts relying more on ChatGPT before the ban or with a background in technology. These findings collectively suggest that the ban reduces analysts' information processing capacities and, therefore, their function as information intermediaries.

Next, we analyze the consequences of the ban on the informational efficiency of the capital market. As a result of reduced analyst forecasts, investors increasingly depend on publicly announced earnings to assess share values. In this case, this shift is accompanied by a widening of bid–ask spreads following the ban, suggesting a decrease in market liquidity and an increase in

information asymmetry. The negative impact on capital market informational efficiency supports the meaningful role of generative AI tools, such as ChatGPT, in enhancing the information processing capacity of capital market participants.

Our results are also subject to several caveats. First, domestic analysts could circumvent such a ban by using proxy servers with foreign IP addresses. As far as we know, OpenAI does not have a policy to block IP addresses used by proxy servers, and sophisticated users, such as coding engineers, who likely benefited the most from ChatGPT's coding capabilities, used proxy servers more extensively and recovered quickly within a few days of the ban being implemented (Kreitmeir and Raschky, 2023). In a counterfactual case, where the ban had been prolonged and extended to other countries, it is possible that a greater share of the user base would have switched to such services. ChatGPT users may also have switched to alternative services during the ban, especially considering that Google's Bard was launched shortly before the ban. However, Bard was initially limited to selecting users on a waitlist in the US and the UK. It was made available to the general public on May 10 and launched in the European Union on July 13, after the ban.

Second, other important events may provide complementary evidence to better measure the economic consequences of AI. For example, OpenAI did not permit individuals to register in China, although it was not fully blocked and was compatible with Mandarin. Throughout February 2023, Chinese authorities took steps to restrict access by ordering social media platforms to remove workarounds and enforce stricter controls on VPN access. The banning of ChatGPT in China is a significant event that could further expand our understanding of the benefits of AI.

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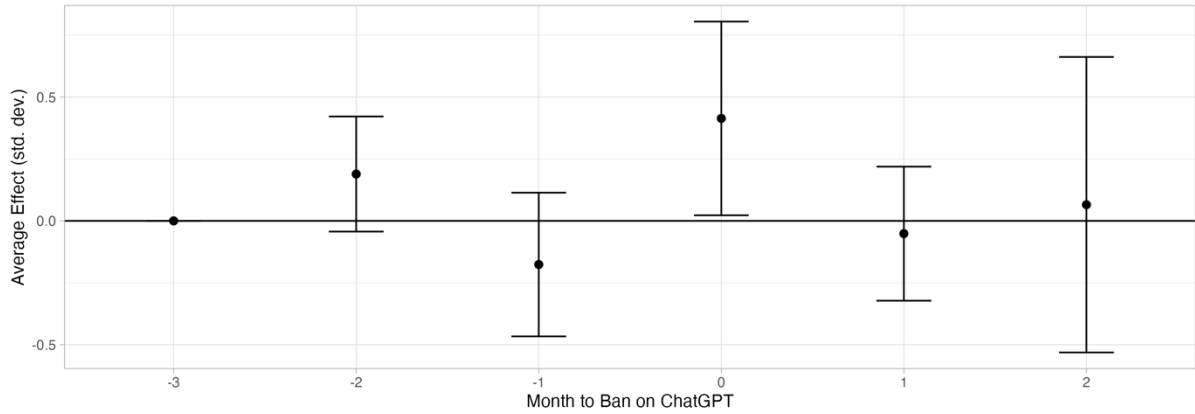
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Figure 1: Dynamic Treatment Effects of the ChatGPT Ban on AI-Generated Contents in Analyst Reports

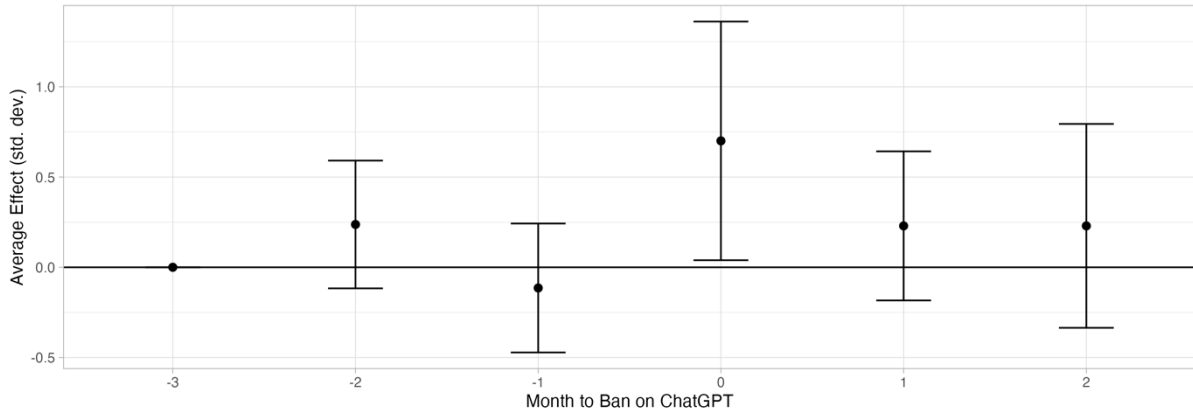
We assess the dynamic treatment effects of the ChatGPT ban on analysts' utilization of AI in drafting analyst reports. Specifically, we estimate the following specification:

$$Text\ Char_{a,t} = \beta_s \sum_{s=-3 \sim +2} Italy\ Analyst_a * D_{s(a,t)} + \beta_3 Italy\ Analyst_a + Controls + \alpha_t + \gamma_n + \epsilon_{a,t}$$

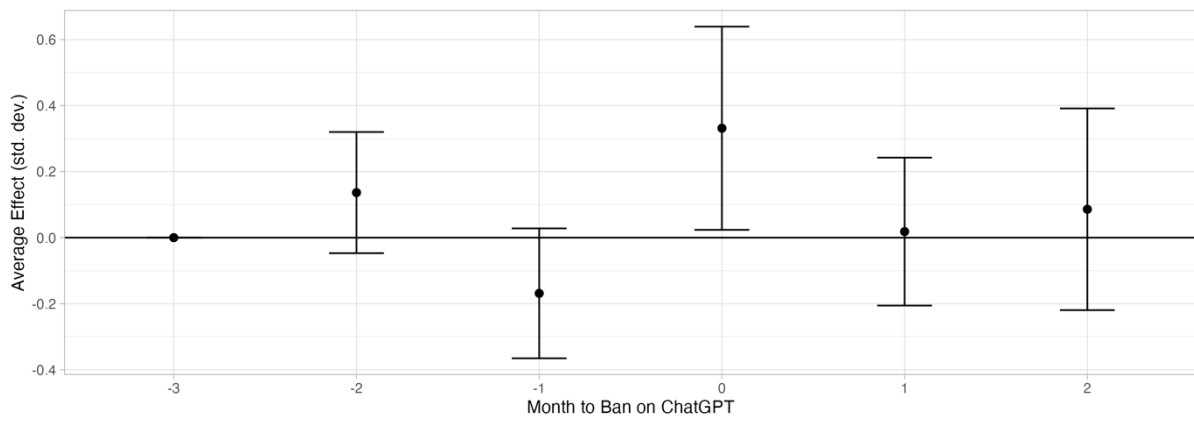
where $Text\ Char_{a,t}$ is the score of AI-generated texts proxied by *perplexity*, *burstiness*, and a composite score of perplexity and burstiness (*Composite Score*) for analyst a in month t . Higher levels of perplexity and burstiness imply a greater likelihood of human writing. The three measures are standardized by calculating the z-score. We report the results for *perplexity*, *burstiness*, and the *Composite Score* in Panels A, B, and C, respectively. *Italy Analyst* is a dummy variable indicating domestic analysts. $D_{s(i,t)}$ is a set of indicator variables that equals one if the time period was s months away from April 2023 ($s = 0$ when ChatGPT was banned), with January 2023 ($s = -3$) omitted and used as the benchmark. Control variables include a set of analysts' characteristics, including work experience, social connection, and education. We add brokerage-house and month fixed effects. Standard errors are clustered at the analyst level.



Panel A: Probability of AI-Generated Content in Analyst Reports – *Perplexity*



Panel B: Probability of AI-Generated Content in Analyst Reports – *Burstiness*



Panel C: Probability of AI-Generated Content in Analyst Reports – *Composite*

Figure 2: Dynamic Treatment Effects of the ChatGPT Ban on Analyst Forecast Frequency

We assess the dynamic treatment effects of the ChatGPT ban on analyst forecast frequency. Empirically, we determine the difference in the changes in forecast frequency among domestic versus foreign analysts using the following specification:

$$Forecast_{i,a,t} = \beta_s \sum_{s=-3 \sim +2} ItalyAnalyst_a * D_{s(a,t)} + \alpha_{i,t} + \gamma_{i,a} + \epsilon_{i,a,t}$$

where $Forecast_{i,a,t}$ is the forecast outcome of an analyst a covering firm i in month t . $ItalyAnalyst$ is a dummy variable indicating domestic analysts, Ban is a time dummy that equals one for the ban period and zero for the pre-periods from January to March 2023, and $D_{s(i,t)}$ is a set of indicator variables that equals one if the time period is s months away from April 2023 ($s = 0$ when ChatGPT was banned), with January 2023 ($s = -3$) omitted and used as the benchmark. We control for firm-by-month and analyst-by-firm fixed effects. Standard errors are clustered by industry.

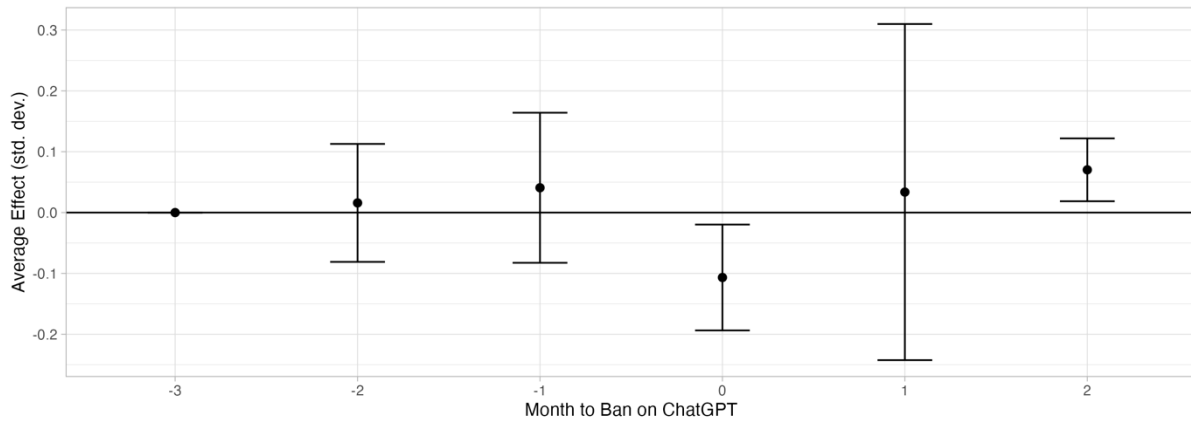
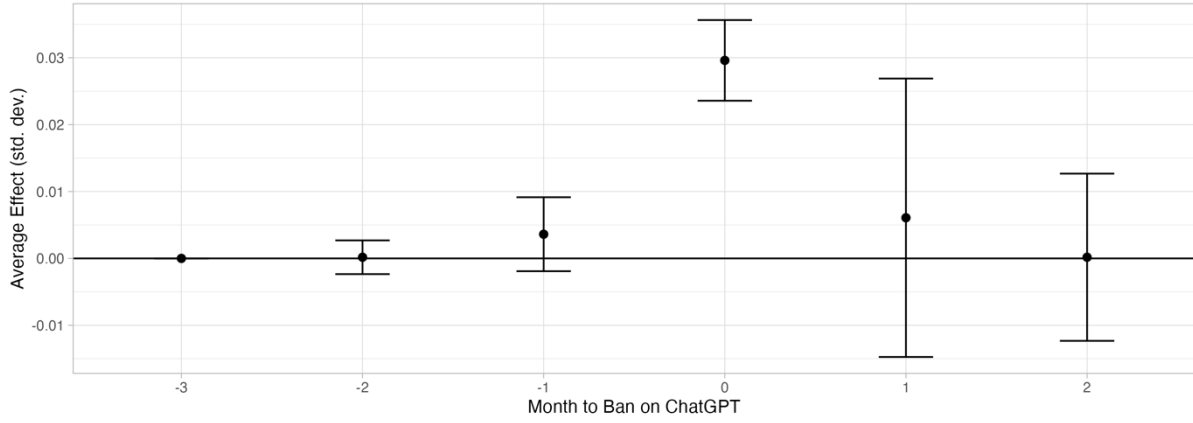


Figure 3: Dynamic Treatment Effects of the ChatGPT Ban on Bid–Ask Spreads

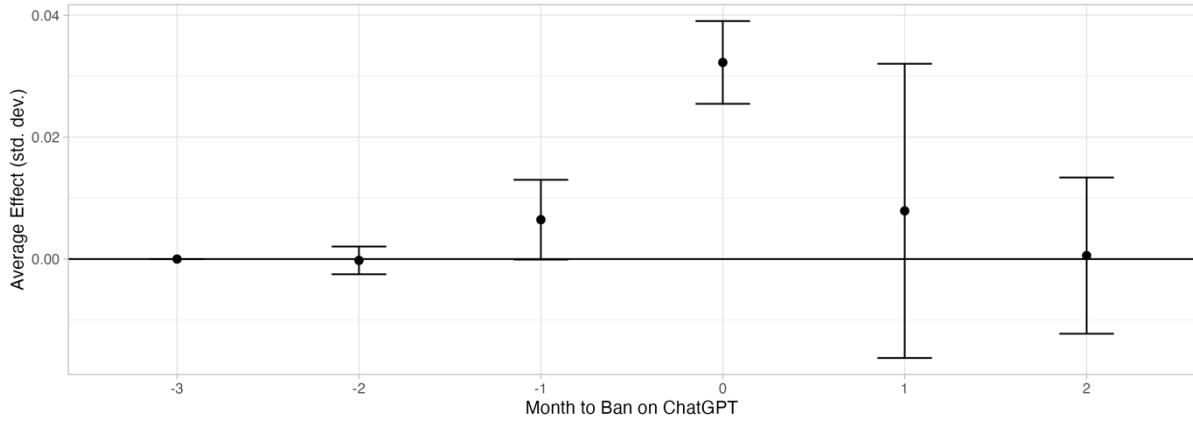
We assess the dynamic treatment effects of the ChatGPT ban on bid–ask spreads. We hypothesize that generative AI could help investors process and summarize unstructured information, leading to better information flow and lower information asymmetry. Empirically, we estimate the following specification:

$$Bid-Ask_{i,t} = \beta_s \sum_{s=-3 \sim +2} Italy_i * D_{s(i,t)} + Controls + \alpha_t + \gamma_i + \epsilon_{i,t},$$

where $Bid-Ask_{i,t}$ is the monthly average of the daily bid–ask spreads, which is the difference between the bid and the ask prices divided by the closing price. $Adj. (Bid-Ask Spreads)$ is the monthly average of the daily bid–ask spreads weighted by the trading volume, and $D_{s(i,t)}$ is a set of indicator variables that equals one if the time period is s months away from April 2023 ($s = 0$ when ChatGPT was banned), with January 2023 ($s = -3$) omitted and used as the benchmark. We control for firm and month fixed effects. Standard errors are clustered at the industry level.



Panel A: Bid–Ask Spreads



Panel B: Adj. Bid–Ask Spreads

Table 1: Summary Statistics

This table presents the summary statistics for the variables used in our main specifications. In particular, it reports the number of observations, means, medians, and standard deviations, as well as the 10th, 25th, 50th, 75th, and 90th percentiles for three samples: the analyst-level sample (characterizing analyst reports), the analyst-firm level sample (focusing on analyst forecasts), and the firm-level sample (analyzing capital market characteristics). See Variable Definition in the Appendix for detailed variable definitions.

Analyst-Level Sample of Analyst Reports								
Variable	Obs.	Mean	St. Dev.	10%	25%	Median	75%	90%
<i>Composite Score</i>	1208	0.049	0.505	-0.403	-0.261	-0.069	0.207	0.630
<i>Perplexity</i>	1208	0.088	1.009	-0.936	-0.545	-0.090	0.496	1.130
<i>Burstiness</i>	1208	0.058	1.062	-0.500	-0.432	-0.242	0.050	0.767
<i>Industry Information</i>	1208	2.338	1.173	1.366	1.654	1.989	2.637	4.021
<i>Tech Major</i>	1208	0.529	0.499	0.000	0.000	1.000	1.000	1.000
<i>Working Years</i>	1208	18.019	8.916	6.000	9.000	16.000	23.000	28.000
<i>Social Connection</i>	1208	437.815	126.992	221.500	447.750	500.000	500.000	500.000
Analyst-Firm-Level Sample of Analyst Forecasts								
<i>Number of Forecasts</i>	2084	0.327	0.528	0.000	0.000	0.000	1.000	1.000
<i>Forecast Dummy</i>	2084	0.299	0.458	0.000	0.000	0.000	1.000	1.000
<i>Forecast Accuracy</i>	362	0.043	0.119	0.001	0.002	0.009	0.031	0.091
<i>Tech Major</i>	2036	0.505	0.500	0.000	0.000	1.000	1.000	1.000
<i>Working Years</i>	2036	11.413	6.982	2.000	5.000	11.000	17.000	22.000
<i>Social Connection</i>	2036	419.796	134.317	197.000	362.000	500.000	500.000	500.000
Firm-Level Sample								
<i>Volatility</i>	1824	2.169	1.349	1.085	1.387	1.844	2.477	3.439
<i>Turnover</i>	1814	19.322	25.014	2.290	5.074	11.351	23.625	41.269
<i>Bid-Ask</i>	1820	0.016	0.019	0.001	0.002	0.010	0.021	0.038
<i>Adj. (Bid-Ask)</i>	1810	0.016	0.020	0.001	0.002	0.009	0.020	0.040
<i>UE</i>	233	0.010	0.045	-0.073	-0.012	0.004	0.024	0.170
<i>Market Cap</i>	1824	2.019	4.776	0.019	0.081	0.315	1.577	5.305
<i>ROA</i>	1824	0.044	0.102	-0.011	0.016	0.051	0.088	0.127

Table 2: Impact of the ChatGPT Ban on the Probability of AI-Generated Content in Analyst Reports

This table reports the results of testing whether the ChatGPT ban affected the analysts' utilization of AI in drafting analyst reports. We adopt two established metrics from the computer science literature that help with AI-generated text detection (AGTD): perplexity and burstiness. Empirically, we employ a difference-in-differences design to estimate whether the probability of using human writing increased for domestic analysts during the ban period compared to their international peers, with the following specification:

$$Text\ Characteristics_{a,t} = \beta_1 Italy\ Analyst_a \times Ban_t + \beta_2 Italy\ Analyst_a + Controls + \alpha_t + \gamma_n + \epsilon_{a,t},$$

where the score of AI-generated texts is proxied by *perplexity*, *burstiness*, and a composite score of perplexity and burstiness (*Composite Score*) for analyst *a* in month *t*. Higher levels of perplexity and burstiness imply a greater likelihood of human writing. *Italy Analyst* is a dummy variable indicating domestic analysts, and *Ban* is a time dummy that equals one for the ban period and zero for the pre-periods from January to March 2023. Control variables include a set of analysts' characteristics, including work experience, social connection, and education. We add brokerage-house and month fixed effects. Standard errors are clustered at the analyst level.

VARIABLES	(1) Perplexity	(2) Burstiness	(3) Composite Score	(4) Perplexity	(5) Burstiness	(6) Composite Score
<i>Italy Analyst</i> × <i>Ban</i>	0.399** (0.190)	0.654** (0.283)	0.351*** (0.110)	0.044 (0.203)	0.397** (0.164)	0.147 (0.091)
<i>Italy Analyst</i> × <i>Ban</i> × <i>Tech Major</i>				0.611** (0.278)	0.398 (0.359)	0.337** (0.141)
<i>Ban</i> × <i>Tech Major</i>				-0.078 (0.084)	0.126 (0.131)	0.016 (0.053)
<i>Italy Analyst</i> × <i>Tech Major</i>				-0.009 (0.148)	0.236 (0.175)	0.075 (0.071)
<i>Italy Analyst</i>	-0.288*** (0.104)	-0.452* (0.244)	-0.247*** (0.085)	-0.280** (0.122)	-0.587** (0.269)	-0.289*** (0.094)
<i>Tech Major</i>				-0.037 (0.057)	-0.119 (0.105)	-0.052 (0.040)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Brokerage FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,208	1,208	1,208	1,208	1,208	1,208
R ²	0.65	0.17	0.40	0.66	0.18	0.41

Table 3: Impact of the ChatGPT Ban on Analyst Forecast Frequency

This table reports the results of testing whether the ChatGPT ban affected analyst forecast frequency. Empirically, we assess the difference in the changes in forecast frequency among domestic versus foreign analysts using the following specification:

$$Forecast_{i,a,t} = \beta_I Italy Analyst_i \times Ban_t + \alpha_{i,t} + \gamma_{i,a} + \epsilon_{i,a,t},$$

where $Forecast_{i,a,t}$ is the forecast outcome of an analyst a covering firm i in month t , $Italy Analyst$ is a dummy variable indicating domestic analysts, and Ban is a time dummy that equals one for the ban period and zero for the pre-periods from January to March 2023. Column 1 controls for firm-level effects, as well as firm, analyst, and month fixed effects. Column 2 (3) controls for a tighter set of fixed effects of analyst- and firm-by-month (firm by month and analyst by firm) fixed effects. Standard errors are clustered by industry.

VARIABLES	(1)	(2)	(3)
		# of Forecast	
<i>Italy Analyst</i> × <i>Ban</i>	-0.125** (0.054)	-0.111** (0.046)	-0.111** (0.046)
Controls	Yes	No	No
Firm FE	Yes	No	No
Month FE	Yes	No	No
Analyst FE	Yes	Yes	No
Firm-by-Month FE	No	Yes	Yes
Analyst-by-Firm FE	No	No	Yes
Observations	2,084	2,084	2,084
R ²	0.21	0.44	0.51

Table 4: Cross-Sectional Tests of Analyst Forecast Frequency

This table reports the results of cross-sectional tests to determine the impact of the ChatGPT ban on analyst forecast frequency. We hypothesize that analysts who were more likely to have adopted ChatGPT in the pre-ban period would be more affected by the ban. Empirically, we employ two proxies for the likelihood of ChatGPT usage: *Ex-Ante Prob (AI-Generated Text)*, which is a dummy variable set to one if the analyst's reports in the pre-ban period were more likely to be written by AI (i.e., the composite score of perplexity and burstiness (*Composite Score*) was lower than the sample median), and *Tech Major*, which indicates whether the analysts had a technical education background. We estimate different specifications with three sets of fixed effects. Columns 1 and 4 control for firm-level controls, as well as firm, analyst, and month fixed effects. Columns 2 and 5 control for analyst- and firm-by-month fixed effects. Columns 3 and 6 control for firm-by-month and analyst-by-firm fixed effects. Standard errors are clustered by industry.

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	# of Forecast					
DUMMY (D) =	<i>Ex-Ante Prob (AI-Generated Text)</i>			<i>Tech Major</i>		
<i>Italy Analyst × Ban × D</i>	-0.203** (0.090)	-0.277** (0.111)	-0.277** (0.111)	-0.371** (0.166)	-0.437** (0.190)	-0.437** (0.190)
<i>Italy Analyst × Ban</i>	0.053 (0.075)	0.120 (0.099)	0.120 (0.099)	0.092 (0.127)	0.118 (0.123)	0.118 (0.123)
<i>Ban × D</i>	0.006 (0.095)	-0.045 (0.053)	-0.045 (0.053)	0.093 (0.108)	0.168 (0.145)	0.168 (0.145)
Controls	Yes	No	No	Yes	No	No
Firm FE	Yes	No	No	Yes	No	No
Month FE	Yes	No	No	Yes	No	No
Analyst FE	Yes	Yes	No	Yes	Yes	No
Firm-by-Month FE	No	Yes	Yes	No	Yes	Yes
Analyst-by-Firm FE	No	No	Yes	No	No	Yes
Observations	944	944	944	2,036	2,036	2,036
R ²	0.24	0.53	0.59	0.23	0.45	0.52

Table 5: Do Foreign Analysts Take Up the Slack?

This table displays the results for tests of whether foreign analysts took up the slack and compensated for domestic analysts' reduced information production. Column 1 shows the changes in the total number of forecasts by both domestic and foreign analysts received by Italian firms from before to during the ban. Column 2 (5) shows the changes in the total number of forecasts by domestic (foreign) analysts. Columns 3 and 4 (6 and 7) further partition the domestic (foreign) analysts by the ex-ante probabilities of AI-generated content appearing in their reports before the ban and their educational backgrounds. *Ex-Ante Prob (AI-Generated Text)* is a dummy variable set to one if the analysts' reports in the pre-ban period were more likely to be written by AI (i.e., the composite score of perplexity and burstiness (*Composite Score*) was lower than the sample median). *Tech Major* indicates whether the analysts had a technical education background. Standard errors are clustered by industry.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
VARIABLES				# of Forecasts			
DUMMY (D)			<i>Ex-Ante Prob</i> <i>(AI-Generated</i> <i>Text)</i>	<i>Tech</i> <i>Major</i>		<i>Ex-Ante Prob</i> <i>(AI-Generated</i> <i>Text)</i>	<i>Tech</i> <i>Major</i>
=							
Sample	All Analysts		Domestic Analysts			Foreign Analysts	
Ban	-0.058 (0.046)	-0.100** (0.036)			0.005 (0.074)		
Ban × D			-0.235*** (0.049)	-0.276*** (0.087)		0.021 (0.071)	0.138 (0.151)
Analyst-by-Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm-by-Month FE	No	No	Yes	Yes	No	Yes	Yes
Observations	2,084	1,252	628	1,204	832	316	832
R ²	0.26	0.24	0.70	0.62	0.27	0.65	0.54

Table 6: Effects of ChatGPT Ban on Analyst Forecast Accuracy

This table reports the effects of the ChatGPT ban on analyst forecast accuracy. Specifically, we assess how forecast accuracy changed from before to during the ban for domestic versus foreign analysts using the following specifications:

$$Forecast\ Accuracy_{i,a,t} = \beta_1 Italy\ Analyst_a \times Ban_t + \alpha_t + \gamma_{i,a} + \epsilon_{i,a,t},$$

where $Forecast\ Accuracy_{i,a,t}$ is the absolute forecast error for analyst a covering firm i in month t , calculated as the absolute value of the difference between the actual earnings per share and the forecasted value of EPS, then deflated by the stock price two days before the announcement of the earnings forecasts. We control for a vector of lagged firm-level characteristics: book-to-market ratio, idiosyncratic volatility, market capitalization, and stock return over the last month. *Italy Analyst* is a dummy variable indicating domestic analysts, and *Ban* is a time dummy that equals one for the ban period and zero for the pre-periods (i.e., January to March). We employ two proxies for the likelihood of ChatGPT usage: *Ex-Ante Prob (AI-Generated Text)*, which is a dummy variable set to one if the analyst's reports in the pre-ban period were more likely to be written by AI (i.e., the composite score of perplexity and burstiness [*Composite Score*] was lower than the sample median), and *Tech Major*, which indicates whether the analysts had a technical education background. We further include analyst-by-firm and month fixed effects. Standard errors are clustered by industry.

VARIABLES	(1)	(2)	(3)
		Absolute Forecast Error (*100)	
DUMMY (D) =		<i>Ex-Ante Prob (AI-Generated Text)</i>	<i>Tech Major</i>
<i>Italy Analyst</i> × <i>Ban</i>	0.946*** (0.040)	0.391 (0.224)	0.919*** (0.051)
<i>Italy Analyst</i> × <i>Ban</i> × <i>D</i>		0.269*** (0.077)	0.219** (0.088)
<i>Ban</i> × <i>D</i>		-0.069** (0.022)	-0.216*** (0.033)
Controls	Yes	Yes	Yes
Analyst-by-Firm FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
Observations	362	149	348
R ²	0.90	0.89	0.90

Table 7: Effects of ChatGPT Ban on Analysts' Processing of Industry-Related Information

This table reports the effects of the ChatGPT ban on analysts' processing of industry-related information. If the ChatGPT ban negatively affected the analysts' information processing capacity, the analysts may have rationally allocated more attention to processing industry-related information as opposed to firm-specific information. Empirically, we rely on the frequency of industry-related words in analyst reports. We first select "industry," "macroeconomy," and "market" as seed words. Second, we employ the word embedding model, "word2vec," to learn all the similar words in the analyst reports. Third, we select a comprehensive industry-related word list based on the output of the word embedding model. *Industry Information* for all reports issued by analyst a in month t is defined as follows:

$$Industry\ Information_{a,t} = \sum_{r=1}^{N_{a,t}} \frac{Industry\ Words_{r,a,t}}{Length_{r,a,t}} / N_{a,t} ,$$

where $Industry\ Words_{r,a,t}$ is the number of all industry-related words in report r issued by analyst a in month t , and $Length_{r,a,t}$ is the number of words in report r by analyst a in month t . $N_{a,t}$ is the number of reports issued by analyst a in month t . We multiply *Industry Information* by 100 to express it as a percentage. We control for a vector of analyst-level characteristics: work experience, social connection, and the highest degree obtained. We further include brokerage-house and month fixed effects. Standard errors are clustered by analyst.

VARIABLES	(1)	(2)	(3)
DUMMY (D) =		Industry Information Ex-Ante Prob (AI-Generated Text)	Tech Major
<i>Italy Analyst</i> × <i>Ban</i>	0.197* (0.109)	-0.0002 (0.080)	0.060 (0.132)
<i>Italy Analyst</i> × <i>Ban</i> × <i>D</i>		0.387** (0.181)	0.441** (0.213)
<i>Italy Analyst</i>	-0.065 (0.069)	-0.034 (0.075)	0.037 (0.077)
<i>Ban</i> × <i>D</i>		-0.120** (0.057)	-0.440*** (0.059)
<i>Italy Analyst</i> × <i>D</i>		-0.047 (0.077)	-0.331*** (0.090)
<i>D</i>		0.012 (0.046)	0.220*** (0.054)
Controls	Yes	Yes	Yes
Brokerage FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
Observations	1,208	1,155	1,208
R ²	0.63	0.62	0.64

Table 8: Impact of the ChatGPT Ban on Market Reaction to Earnings Announcements

This table reports the results of testing whether the ChatGPT ban affected the stock market's reaction to earnings announcements. We hypothesize that if the ban reduced analysts' production of forward-looking information before earnings announcements, it may have increased investors' reactions to earnings announcements. Empirically, we estimate the following regression:

$$Ret_{i,t} = \beta_1 Italy_i \times Ban_t \times UE_{i,t} + \beta_2 Italy_i \times Ban_t + \beta_3 Italy_i \times UE_{i,t} + \beta_4 Ban_t \times UE_{i,t} + \beta_5 UE_{i,t} + \beta_6 Italy_i + \beta_7 Controls \times UE_{i,t} + Controls + \alpha_t + \gamma_j + \epsilon_{i,t},$$

where $Ret_{i,t}$ is the cumulative abnormal return for firm i in four different time windows: one, three, five, and seven days centered around the earnings announcement date t . Ban is a time dummy that equals one for the ban period and zero for the pre-periods. The unexpected earnings ($UE_{i,t}$) are calculated as the unexpected part of the announced earnings scaled by the stock price. We include firm-level controls: *Loss*, *Size*, *Market-to-Book*, *Leverage*, *Earnings Persistence*, and *Beta*. We control for industry and month fixed effects and the interaction between the control variables and UE . We cluster the standard errors by industry.

VARIABLES	(1) Ret [0]	(2) Ret [-1, +1]	(3) Ret [-2, +2]	(4) Ret [-3, +3]
<i>Italy × Ban × UE</i>	0.829*** (0.154)	0.557* (0.296)	0.610** (0.259)	0.826*** (0.239)
<i>Ban × UE</i>	-0.668*** (0.130)	-0.678*** (0.165)	-0.480** (0.173)	-0.536** (0.200)
<i>Italy × UE</i>	-0.369*** (0.099)	-0.496*** (0.143)	-0.671*** (0.119)	-0.716*** (0.151)
<i>Italy × Ban</i>	0.011 (0.018)	0.040* (0.020)	0.053** (0.023)	0.053** (0.021)
<i>UE</i>	0.987* (0.529)	1.61*** (0.529)	1.02** (0.468)	1.19** (0.470)
<i>Italy</i>	0.006 (0.007)	0.019* (0.011)	0.021** (0.009)	0.017 (0.014)
Controls	Yes	Yes	Yes	Yes
UE × Controls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
Observations	192	192	192	192
R ²	0.21	0.27	0.27	0.32

Table 9: Impact of the ChatGPT Ban on the Bid–Ask Spreads

This table reports the results of testing whether the ChatGPT ban affected bid–ask spreads. We hypothesize that generative AI could help investors process and summarize unstructured information, leading to better information flow and lower information asymmetry. Empirically, we estimate the following specification:

$$Bid-Ask_{i,t} = \beta_1 Italy_i \times Ban_t + Controls + \alpha_i + \gamma_t + \epsilon_{i,t},$$

where $Bid-Ask_{i,t}$ is the monthly average of the daily bid–ask spreads, which is the difference between the bid and the ask prices divided by the closing price. $Adj. (Bid-Ask Spreads)$ is the monthly average of the daily bid–ask spreads weighted by the trading volume. We control for firm and month fixed effects. Standard errors are clustered at the industry level.

VARIABLES	(1) Bid–Ask Spreads	(2) Adj. (Bid–Ask Spreads)
<i>Italy × Ban</i>	0.028*** (0.002)	0.031*** (0.002)
Controls	Yes	Yes
Firm FE	Yes	Yes
Month FE	Yes	Yes
Observations	1,553	1,543
R ²	0.85	0.80

Table 10: Cross-Sectional Tests on the Bid–Ask Spreads

This table reports the results of cross-sectional tests to determine the impact of the ChatGPT ban on the adjusted bid–ask spreads. Specifically, we conduct cross-sectional tests to shed light on the mechanisms driving the increased information asymmetry. Column 1 shows the results for *Italy* $\times$ *Ban* interacting with an indicator for high analyst coverage, which equals one for firms with higher-than-sample-median analyst coverage before the ChatGPT ban. Similarly, Column 2 (3) shows the results for firms with a higher percentage of foreign investors (institutional investors) than the sample median. We control for firm and month fixed effects. Standard errors are clustered at the industry level.

VARIABLES	(1) Adj. (Bid–Ask Spreads)	(2) Adj. (Bid–Ask Spreads)	(3) Adj. (Bid–Ask Spreads)
<i>DUMMY (D) =</i>	<i>High Analyst Coverage</i>	<i>High Foreign Investor</i>	<i>High Institutional Investor</i>
<i>Italy</i> $\times$ <i>Ban</i> $\times$ <i>D</i>	-0.016*** (0.002)	-0.011*** (0.003)	-0.011*** (0.004)
<i>Italy</i> $\times$ <i>Ban</i>	0.039*** (0.002)	0.036*** (0.002)	0.036*** (0.003)
<i>Ban</i> $\times$ <i>D</i>	0.0003 (0.0005)	0.0010* (0.0005)	0.0008 (0.0007)
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
Observations	1,543	1,467	1,531
R ²	0.81	0.81	0.80

Appendix A: Variable Definitions

This table provides a detailed description of the main variables used in our analysis.

Variable Name	Definition
<i>Ban</i>	A dummy variable set to one for the ban period from March 31 (inclusive) to April 28 (inclusive), covering a total of 21 working days, and zero for the pre-ban and post-ban periods. For comparability across time periods, we defined each time unit in the pre-ban, during-ban, and post-ban periods to include exactly 21 working days. The pre-ban period consists of three consecutive 21-working-day units (January 2 to March 30), referred to as January, February, and March for simplicity. The post-ban period comprises two consecutive 21-working-day units (April 29 to June 27), referred to as May and June.
<i>Italy Analyst</i>	A dummy variable set to one if an analyst was based in Italy and zero otherwise. Analysts are classified as Italy-based using location information based on their office addresses, email domains, or phone numbers sourced from Refinitiv or analyst reports.
<i>Number of Forecasts</i>	The number of forecasts by analyst a covering firm i in month t .
<i>Forecast Dummy</i>	A dummy variable set to one if at least one forecast was issued by analyst a covering firm i in month t , and zero otherwise.
<i>Forecast Accuracy</i>	The absolute value of the difference between the actual and forecasted values of EPS deflated by the stock price two days before the earnings forecasts.
<i>Perplexity</i>	A measure of the procedural nature of AI text generation. The inherent variability and uniqueness of human writing leads to higher perplexity scores. We standardized the measure by calculating the z-score.
<i>Burstiness</i>	A measure of how perplexity varies throughout a document. Greater burstiness tends to indicate that the document is more human-like. We standardized the measure by calculating the z-score.
<i>Composite Score</i>	The average of the z-scores of the two measures: <i>Perplexity</i> and <i>Burstiness</i> .
<i>Industry Information</i>	A measure of the proportion of industry-related content discussed in analyst reports, which was calculated based on the frequency of industry-related words. The construction process was as follows: First, we began by selecting “industry,” “macroeconomy,” and “market” as initial seed words. Second, using the “word2vec” model, we identified all words similar to the seed words across analyst reports. Finally, based on the output of the word embedding model, we compiled a detailed list of industry-related words (see Appendix C for details). Based on this list, we defined <i>Industry Information</i> for all reports issued by analyst a in month t as $Industry\ Information_{a,t} = \sum_{r=1}^{N_{a,t}} \frac{Industry\ Words_{r,a,t}}{Length_{r,a,t}} / N_{a,t},$ where $Industry\ Words_{r,a,t}$ is the number of all industry-related words in report r issued by analyst a in month t, and $Length_{r,a,t}$ is the number of words in report r by analyst a in month t. $N_{a,t}$ is the number of reports issued by analyst a in month t. We multiplied <i>Industry Information</i> by 100 to express it as a percentage.
<i>Tech Major</i>	A dummy variable set to one if the analyst majored in technical subjects as defined by Frank et al. (2023), and zero otherwise.
<i>Working Year</i>	A measure of how many years an analyst had worked in this particular career field.
<i>Social Connection</i>	A measure that represents the number of LinkedIn connections an analyst had, ranging from 0 to 500. Any value above 500 is displayed as “500+” on LinkedIn.
<i>Unexpected Earnings</i>	Similar to Livnat and Mendenhall (2006) and Mian and Sankaraguruswamy (2012), we calculated the unexpected earnings, UE, as follows: $UE_{i,t} = \frac{Earnings_{i,t} - Earnings_{i,t-1}}{P_{i,t}},$

	where $Earnings_{i,t}$ is the earnings per share of firm i at time t , and $P_{i,t}$ is the price per share two days before the earnings announcement date to avoid contamination.
<i>Cumulative Abnormal Return</i>	Cumulative abnormal return was obtained by cumulating the daily abnormal return over a certain period. The daily abnormal return is defined as the difference between the daily stock return and the value-weighted daily market return. Alternatively, the daily abnormal return was also adjusted based on the CAPM model: $R_t^{mkt\ adj} = R_t - \beta_t R_{MKT-RF,t},$ where β_t is estimated by running the following regression with daily returns ranging from [-252, -1]: $R_t - R_t^f = \alpha_t + \beta_t R_{MKT-RF,t} + \varepsilon_t$ where R_t^f is the risk-free rate, and $R_{MKT-RF,t}$ is the value-weighted portfolio return of all stocks in the EU plus the UK in excess of the risk-free rate.
<i>Return</i>	The monthly buy-and-hold stock return.
<i>Volatility</i>	The monthly standard deviation of the daily raw return.
<i>Idio Vol</i>	The volatility of the daily abnormal return calculated based on the CAPM model over a month.
<i>Turnover</i>	The monthly average of the ratio of traded shares to the total shares outstanding, multiplied by 100.
<i>Bid-Ask</i>	The monthly average of daily bid-ask spreads, which is the bid price minus the ask price divided by the closing price.
<i>Adj. (Bid-Ask)</i>	The monthly average of daily bid-ask spreads weighted by daily trading volume.
<i>Book-to-Market Ratio</i>	The book value of equity per share divided by the stock price at the month's end.
<i>Market Cap</i>	The market capitalization in EUR billions at the end of a month.
<i>ROA</i>	Earnings before interests and taxes (EBIT) divided by total assets.

Appendix B: Testing the Effectiveness of the Perplexity- and Burstiness-Based Algorithms

We conduct a laboratory experiment to test the effectiveness of the perplexity- and burstiness-based algorithms in detecting AI-generated texts. Approved by our institution’s Institutional Review Board (IRB), the experiment takes place in a computer lab at a major research university, ensuring consistent technological and environmental conditions. We recruit 35 participants with backgrounds in accounting and finance through university email broadcasts targeting senior and graduate students in these departments. Eligibility requires the completion of at least two courses in financial analysis or related subjects to ensure content familiarity. Participants, most of whom are non-native English speakers, are instructed to write equity reports in English, which mirrors the scenario in which domestic analysts often produce such reports in a non-native language. We do not collect personal information, and each participant signs a consent form outlining the data usage policy, which ensures confidentiality and IRB compliance.

Each participant receives a packet containing the earnings call transcript, the 2022 performance statistics of Prada S.p.A. (PRDSY), a glossary of financial terms, and detailed instructions for the report requirements. They are required to write their reports on university-provided computers using standard text processing software. We provide the following instructions: “Please write a short report based on the earnings call transcript. Your report will answer two questions. First, what are Prada’s 2022 performance and driving factors? You shall discuss at least three aspects: Prada products, Miu Miu products, and financial metrics, such as revenue, margin, etc. Second, you shall focus on the discussions about the Chinese market, including what happened in 2022 Q4 (COVID restrictions) and the outlook for 2023 (reopening).” These questions are open-ended yet specific enough to guide the analysis, while allowing for individual interpretation and insight. Reports are submitted electronically via a secure university platform that timestamped each submission to ensure adherence to deadlines. All reports are coded with participant IDs instead of names to maintain anonymity during the analysis phase. We utilize these human-written reports to conduct a verification test. First, we use ChatGPT 3.5 to polish the human-written reports. Second, we ask ChatGPT 3.5 to summarize the raw earnings call transcript using the following prompt: “You are a financial analyst. You will be given the Prada Group’s earnings call transcript. Please summarize Prada’s 2022 performance and key drivers. Please also summarize the discussions on China. Your answer should be readable to investors.” We then insert the content from the earnings call transcript into ChatGPT 3.5.

The table below presents our results comparing human-written reports to those polished or generated by ChatGPT 3.5 using perplexity and burstiness measures. Column 1 shows the average perplexity and burstiness for the human-written texts, while Column 2 shows these results for texts polished by ChatGPT, and Column 3 for texts entirely generated by ChatGPT. Standard deviations are in parentheses, with AI-written samples exhibiting the lowest variability. We use Welch *t*-tests to assess the significance of differences among the three types of reports, as this test accommodates differing standard deviations across samples—a suitable choice given the expected variability due to distinct writing practices. The *t*-test reveals that the difference between human and AI-written samples (Column 7) has a higher *t*-statistic than that between human and AI-polished samples (Column 5).

Our findings indicate that the perplexity and burstiness scores are significantly higher for human-written reports compared to both AI-polished and AI-written reports. Perplexity (burstiness) decreases by 14.20% (25.48%) when comparing human-written to AI-polished reports, and by 16.67% (35.0%) when comparing human-written to AI-written reports.

	(1) Human Written	(2) AI Polished	(3) AI Written	(4) Difference (1)-(2)	(5) <i>t</i> -stat	(6) Difference (1)-(3)	(7) <i>t</i> -stat
Perplexity	167.80 (47.72)	143.97 (33.23)	139.83 (25.20)	23.83	2.42	27.97	3.07
Burstiness	4284.06 (2516.56)	3192.66 (2795.59)	2782.77 (915.56)	1091.40	1.72	1501.29	3.31

Appendix C: Expanded Dictionary of Industry-Related Keywords

We report an expanded dictionary of industry-related keywords. We construct the dictionary through the following steps. First, we select “industry,” “macroeconomy,” and “market” as seed words. Second, we employ the word embedding model, “word2vec,” to learn all the similar words in the analyst reports. Third, we select a comprehensive industry-related word list based on the output of the word embedding model.

keywords	keywords	keywords
economic	industrial	lockdown
market	e_commerce	society
economy	globally	eurozone
industry	competitiveness	energy_crisis
macro	demographic	macroeconomic_condition
sector	interest_rate_rise	market_environment
demand	macroeconomic_context	geopolitical_tension
supply_chain	macro_trend	domestic
macro_environment	war	regional
region	gdp_growth	economic_recovery
competition	government	climate
country	core_inflation	global_economy
Economic	covid	rival
Geopolitical	business_environment	competitive
market_growth	policy	new_entrant
emerge_market	national	intense_competition
economic_cycle	global_economic	competitive_pressure
market_demand	regulation	domestically
Pandemic	population	supply_chain_constraint
Macroeconomic	market_volatility	cost_live_crisis
Global	demand_trend	weather_condition
Inflation	domestic_market	covid_19
market_trend	rise_interest_rate	regulatory_landscape
economic_environment	price_competition	post-pandemic
World	peer	post-covid
Nation	pricing_environment	geopolitical_crisis
macroeconomic_environment	interest_rate	economic_slowdown
competitive_environment	regulatory_environment	macro-economic
Inflationary	new_entrant	

Online Appendix

Table OA1: Covariate Balances in the Pre-Ban Period for Treated and Control Samples

This table presents the covariate balances for the variables used in our main specifications. It provides the results of two-sample *t*-tests comparing the means of the treated and control groups during the pre-ban period. We compare domestic analysts with foreign analysts who covered Italian firms in the analyst-level and analyst-firm-level samples. For the firm-level sample, we compare Italian firms with matched non-Italian firms. Each Italian firm is matched with a non-Italian firm in the same industry, with the closest market capitalization and ROA, and without replacement.

Analyst-Level Sample for Analyst Reports				
Variable	Domestic Analyst	Foreign Analyst	Difference	t-stats
<i>Composite Score</i>	0.186	0.166	0.019	0.200
<i>Perplexity</i>	0.574	0.246	0.328	1.200
<i>Burstiness</i>	0.110	0.083	0.028	0.345
<i>Industry Information</i>	2.146	2.250	-0.104	-1.192
<i>Tech Major</i>	0.538	0.517	0.022	0.206
<i>Working Years</i>	19.013	17.788	1.225	0.676
<i>Social Connection</i>	423.577	439.359	-15.782	-0.622
Analyst-Firm-Level Sample for Analyst Forecasts				
<i>Number of Forecasts</i>	0.359	0.389	-0.030	-0.574
<i>Forecast Dummy</i>	0.358	0.384	-0.026	-0.508
<i>Forecast Accuracy</i>	0.020	0.028	-0.008	0.020
<i>Tech Major</i>	0.475	0.517	-0.042	-0.318
<i>Working Years</i>	13.542	11.250	2.292	1.383
<i>Social Connection</i>	383.549	455.306	-71.757	-1.896
Firm-Level Sample Matched by PSM				
Variable	Domestic Firm	Matched EU/UK Firm	Difference	t-stats
<i>Mkt Cap</i>	2258.04	1974.96	283.09	0.73
<i>ROA</i>	0.04	0.04	0	0.94

Table OA2: Effects of the ChatGPT Ban on Analyst Forecast Frequency Dummy

This table reports the effects of the ChatGPT Ban on analyst forecast frequency. The dependent variable is a dummy variable set to one if at least one forecast was issued by an analyst a covering firm i in month t , and zero otherwise. *Italy Analyst* is a dummy variable indicating domestic analysts, and *Ban* is a time dummy that equals one for the ban period and zero for the pre-periods (i.e., January to March). We compare the outcomes for the analysts in Italy with those in other countries covering the same Italian firm by controlling for firm-by-month fixed effects. We further include analyst-by-firm fixed effects. Standard errors are clustered by industry.

VARIABLES	(1)	(2)	(3)
<i>DUMMY (D) =</i>		Forecast Dummy Ex-Ante Prob (AI- Generated Text)	Tech Major
<i>Italy Analyst</i> × <i>Ban</i>	-0.082* (0.043)	0.145* (0.079)	0.118 (0.106)
<i>Italy Analyst</i> × <i>Ban</i> × <i>D</i>		-0.303** (0.110)	-0.391** (0.178)
<i>Ban</i> × <i>D</i>		0.009 (0.045)	0.166 (0.122)
Analyst-by-Firm FE	Yes	Yes	Yes
Firm-by-Month FE	Yes	Yes	Yes
Observations	2,084	944	2,036
R ²	0.51	0.59	0.52

Table OA3: Placebo Tests for the Effects of the ChatGPT Ban on Analyst Forecasts

This table reports the effects of the ChatGPT Ban on analyst forecast frequency. The dependent variable is a dummy variable set to one if at least one forecast was issued by an analyst a covering firm i in month t , and zero otherwise. *Italy Analyst* is a dummy variable indicating domestic analysts, and *Ban* is a time dummy that equals one for the ban period and zero for the pre-periods (i.e., January to March). *Low Social Connection* is a dummy variable set to one if the number of connections of an analyst on LinkedIn was below the sample median and zero otherwise. *Low Experience* is a dummy variable set to one if the career length of an analyst was below the sample median and zero otherwise. We compare the outcomes for the analysts in Italy with those in other countries covering the same Italian firm by controlling for firm-by-month fixed effects. We further include analyst-by-firm fixed effects. Standard errors are clustered by industry.

VARIABLES	(1) # of Forecast	(2) Forecast Dummy	(3) # of Forecast	(4) Forecast Dummy
<i>DUMMY (D) =</i>	<i>Low Social Connection</i>		<i>Low Experience</i>	
<i>Italy Analyst</i> × <i>Ban</i> × <i>D</i>	-0.033 (0.166)	-0.041 (0.142)	0.025 (0.136)	0.013 (0.105)
<i>Italy Analyst</i> × <i>Ban</i>	-0.083 (0.097)	-0.052 (0.077)	-0.111** (0.050)	-0.073 (0.044)
<i>Ban</i> × <i>D</i>	-0.022 (0.154)	-0.013 (0.132)	-0.023 (0.096)	0.007 (0.073)
Analyst-by-Firm FE	Yes	Yes	Yes	Yes
Firm-by-Month FE	Yes	Yes	Yes	Yes
Observations	2,036	2,036	2,036	2,036
R ²	0.52	0.52	0.52	0.52

Table OA4: Effects of the ChatGPT Ban on Stock Return Volatility and Turnover

This table reports the effects of the ChatGPT Ban on stock return volatility and turnover. *Volatility* is a monthly measure of the standard deviation of daily raw return. *Idio_Vol* is the standard deviation of daily abnormal returns based on the CAPM model, and *Turnover* is the monthly average of the ratio of traded shares to the total shares outstanding. We control for firm and month fixed effects. Standard errors are clustered by industry.

VARIABLES	(1) Volatility	(2) Idio_Vol	(3) Turnover
<i>Italy × Ban</i>	-0.087 (0.112)	-0.097 (0.090)	-3.35** (1.40)
<i>Book-to-Market</i>	-0.061*** (0.008)	-0.070*** (0.009)	-0.136 (0.093)
<i>Idio Vol</i>	-0.049 (0.055)	-0.061 (0.061)	-0.614 (0.910)
<i>Market Cap</i>	0.648 (0.530)	0.460 (0.470)	18.2** (6.69)
<i>Return</i>	0.585 (0.397)	0.359 (0.392)	11.5 (11.1)
Firm FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
Observations	1,557	1,417	1,547
R ²	0.59	0.59	0.77

Online Appendix

Writing Samples from the Experiment

[Human-Only Writing Sample]:

2022 Performance and the Driving Factors

Sharp increased net revenue, 21% for total of EUR4.2 billion. The reason for this is the great identity and the appeal of Prada itself, the brand image really increases the sales. The products successful design also boosting selling, there are many well- performed categories. Profitability also increased well, major growth in all gross margin and EBIT. The increased profitability can be explained by the full-price sales, the products are been sold at their fully prices without any discount. This driven factors for full price sales may be the good store experiences for customers since Prada has conducted 100 store renovation.

China discussion: The covid lockdown of China really dragged the mainland China sales for 2022 year ended, but regarding to the reopening for Chinese government, the revenue may growth starting from beginning of 2023. The main discussion may involve what the main focus, is it to attract new customers to shop or motivates old customers to shop again at the store. The huge marketing expense may take into consideration if Prada chose to go the first way but with good potential, the old existing customers were shopping online during the lock-down period so keep trying effort on them may not increase the revenue that much.

However, regrading to the economic recession within China for recent years, the main way for Parda may be changing, not dreaming on fully price policy anymore and try some discount boosting sale policies.

[AI-Polished Writing Sample]:

2022 Performance Highlights

Net Revenue Growth: Prada's sharp increase in net revenue by 21% to EUR 4.2 billion is significant and underscores the strength of the brand and its global appeal. This growth is attributed to the brand's identity, appealing designs, and diverse, well-received product categories. The ability of Prada to maintain its brand image and continuously innovate in design has played a crucial role in attracting customers and driving sales.

Increased Profitability: The rise in profitability, reflected in the growth of both gross margin and EBIT, can largely be credited to the strategy of selling products at full price. This approach has been facilitated by an improved store experience, notably through the renovation of 100 stores, which enhances customer satisfaction and loyalty. Selling at full price also suggests that Prada has successfully positioned its products as premium offerings that customers are willing to purchase without discounts, highlighting the brand's strong market positioning and product desirability.

The China Discussion: Impact of Covid Lockdowns: The lockdowns in China due to COVID-19 significantly impacted sales in mainland China for the year 2022. However, the reopening of the country as announced by the Chinese government presents an opportunity for revenue growth from the beginning of 2023. The challenge here is to strategize effectively to capture this potential growth, which could involve attracting new customers or re-engaging existing ones.

Strategic Considerations: The decision between focusing on attracting new customers versus encouraging repeat purchases from existing customers involves weighing the cost of extensive marketing efforts against the potential for increased sales. Given that existing customers have demonstrated a willingness to shop online during lockdowns, efforts to retain and enhance online shopping experiences could also be crucial.

Economic Recession and Pricing Strategies: The economic recession in China poses additional challenges. It may necessitate a reevaluation of the full-price policy in favor of introducing strategic

discounts or promotional sales to boost purchases. This shift could help counteract the reduced spending power and maintain sales volumes, albeit at potentially lower margins.

In summary, while Prada's performance in 2022 was impressive, the brand faces a complex landscape in China, influenced by post-COVID recovery, economic challenges, and consumer behavior shifts. Adapting to these conditions by balancing brand image, pricing strategies, and customer engagement will be essential for sustaining growth and profitability.

[AI-Only Writing Sample]:

Prada Group's Performance in 2022 and Outlook on China

2022 Performance Highlights

The Prada Group demonstrated impressive resilience and growth in 2022 amidst challenging global conditions. Key achievements include:

Brand Rankings and Recognition: Prada and Miu Miu emerged as leading luxury brands, with Prada ranked first and Miu Miu fourth among top luxury brands. Miu Miu was also named The Brand of the Year, underscoring the strong identity and appeal of the Group's brands.

Financial Growth: The Group reported a 21% increase in net revenues, reaching EUR 4.2 billion. This growth was powered by a 24% sales increase in the retail channel across all product categories, totaling EUR 3.7 billion.

Profitability: Significant improvements in profitability were noted, with a gross margin of 78.8% and an adjusted EBIT margin of 20.1%, resulting in EUR 845 million. The net financial position was positive at EUR 535 million.

Strategic Initiatives: Over 100 store renovation projects were completed, enhancing merchandising and client-driven activities. Notable product launches included the Eternal Gold jewelry collection and the Prada Paradoxe fragrance.

Product and Market Growth: All product categories experienced double-digit growth. Miu Miu particularly benefited from increased brand awareness and appreciation, with retail revenues growing by 20% and reaching net revenues of EUR 489 million.

Discussions on China: China posed a unique challenge in 2022 due to COVID restrictions, impacting sales. However, the Group's performance in other regions offset this impact. Discussions about China included:

Market Recovery: Positive indicators were observed in early 2023, with an excellent Chinese New Year and the resumption of internal travel, contributing to new dynamism in retail stores.

Investment Plans: The Group plans to increase marketing and store investments in China, viewing it as a key priority market. Different scenarios are being considered to adapt to market changes, ensuring readiness for any situation.

Consumer Dynamics: The reopening of China is attracting more new customers than repeat customers to Prada stores, indicating a strong brand appeal and a successful journey in enhancing brand equity over the past three years.

Investor Takeaways: The Prada Group's strategic execution and brand appeal led to significant growth in 2022, overcoming global challenges. While China presented specific challenges due to COVID-19 restrictions, early signs of recovery in 2023 offer optimism. Increased investments and a focus on attracting new customers in China highlight the Group's adaptive strategies and commitment to maintaining its strong market position. Investors should view these developments as positive indicators of the Group's resilience, strategic foresight, and potential for sustained growth.